

Impacts of US deforestation on rainfall from mesoscale convective systems

by

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The student author, whose presentation of the scholarship herein was approved by the program of study committee, is solely responsible for the content of this thesis. The Graduate College will ensure this thesis is globally accessible and will not permit alterations after a degree is conferred.

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NOMENCLATURE

Term	Definition of the term
CAM	Convective Allowing Model
CAPE	Convective Available Potential Temperature
CESM	Community Earth System Model
CESM PFT	Community Earth System Model Plant
CIN	Convective Inhibition
CLM	Community Land Model
CONUS	Contiguous United States
CPC	Climate Prediction Center
ELAI	Effective Leaf Area Index (Product of Fractional Vegetation and Leaf Area Index)
FVEG	Fractional Vegetation
GPLLJ	Great Plains Low Level Jet
LAI	Leaf Area Index
LLJ	Low Level Jet
LULC	Land Use Land Cover
LULC change	Land Use Land Cover Change
MCS	Mesoscale Convective System
NOAA PSL	National Oceanic and Atmospheric Association Physical Sciences Laboratory
PBL	Planetary Boundary Layer
WRF	Weather Research and Forecasting Model

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ABSTRACT

This study investigates the effects of land use land cover (LULC) change on the 1993 and 2008 US Midwest flood events using the Weather Research and Forecasting model with the Noah-MP land surface model. Deforestation in the United States has led to large changes in LULC since the year 1850, in which forests have been replaced by croplands. This study examines the overall changes to the atmosphere caused by changing LULC from the present-day to the 1850 vegetation cover, and particularly the response of precipitation amounts and extremes. The LULC experiments caused a change in surface latent heat flux through the change in effective leaf area index. In the Noah-MP model, the latent heat flux over agricultural land was often greater than that over forests. Surface sensible heat flux and temperature increased with increasing forest cover in the 1850 experiments due to a lower albedo of forests. The increased temperatures in the 1850-vegetation experiment led to a decrease in pressure gradients across the domain, leading to a weakened great plains low level jet. The weakened low level jet weakened the dynamical forcing along the front and decreased the transport of warm, moist air. The decreased surface latent heat flux in the 1850-vegetation experiment led to decreased available moisture at the surface. It was also found that precipitable water decreased on average across the United States with 1850-vegetation. These changes led to a decrease in overall precipitation in the 1850-vegetation experiments. Although precipitation decreased, the atmospheric supply of water at the surface (precipitation minus evaporation) did not significantly change due to a decrease in surface evaporation in the 1850-vegetation experiment. This essentially offsets the change in precipitation.

CHAPTER 1. GENERAL INTRODUCTION

1.1 Historical land-use change in the US since the 1800s

Land cover refers to the quantitative state of the land surface, for example the area of ground covered by vegetation, while land use is the human use of the land. A few examples of land use are croplands, forests, waterbodies, and urban spaces. Land-use and land cover (LULC) in North America changed drastically during the early industrialization of the United States. Significant changes occurred in the colonial period as settlers expanded agricultural land in the United States. This accelerated during the industrial boom and westward expansion in the United States. Historically, the clearing of US forests occurred over the eastern half of the United States and has shifted from the eastern to the central part of the US over the past several decades (Ramankutty et al., 2010). It may be noted that changes in LULC across the United States also include considerable changes in wetland distribution between 1780 and 1980. Arkansas, Illinois, Indiana, Missouri, and other states have lost 70% or more of their wetland acreage (Dahl 1990).

Reforestation is proposed as a future ‘natural climate solution’ for combating climate change. It has been stated that reforesting a proposed 146 million acres with 75.3 billion trees could capture 370 million tons of carbon dioxide yearly (Cook-Patton 2023). The estimated reforestation area is large (Bastin et al., 2019); roughly half of the US acreage that is thought to have been lost since European colonization (Ramankutty, Heller, and Rhemtulla 2010), but is located outside of existing agricultural regions. If past deforestation studies are used as a guide to the effects of reforestation, these reforestation efforts will affect surface friction, planetary boundary layer (PBL) depth, and the low-level jet (Seth and Giorgi, 1998). 37% of the mitigation needed between now and 2030 to keep global temperature rise below 2% could come from

natural climate solutions such as reforestation. Reforesting with approximately 7.62 billion trees could capture 535 million tonnes of CO₂ annually (nature.org). This is the equivalent of removing 116 million cars from the road.

LULC has biogeochemical and biophysical effects on the local and global scale. Many biophysical processes, such as latent heat flux, sensible heat flux, and albedo, affect the PBL on the order of days to months (DeLucia 2015). Latent heat is the energy absorbed or released during the phase change of a substance (water). Evaporation requires energy to go into latent heat, which causes the surface to cool. Latent heat is released by vegetation in evapotranspiration or soils that hold water. Latent heat flux may change with vegetation and soil type. Sensible heat flux is the flow of energy that causes a change in temperature. An example of sensible heat flux is the ground heating during the day and transferring this energy upward to heat the atmosphere in turbulent eddies. Albedo is the proportion of shortwave radiation reflected by the surface. Lighter surfaces, such as snow, have a higher albedo. Albedo can affect sensible heat flux because a darker surface absorbs more energy from the sun. A change in vegetation type may change the albedo of the surface. On a global scale, there are biogeochemistry processes between the land cover and the climate, such as the exchange of greenhouse gasses (DeLucia 2015). These take place on the order of decades and centuries. These processes are not necessarily relevant in this study because we investigate the changes on a time scale of hours to months.

This research is relevant to many aspects of Earth Science research. Reforestation will likely increase as governments and corporations incentivize tree planting as a climate change solution. Model simulations of past deforestation can be used to understand the implications of ongoing reforestation efforts, which may impact extreme precipitation during floods. Model simulations can also help us understand the implications of expanding agriculture across the

United States. This study will examine past events (the 1993 and 2008 Midwest US floods) to investigate how historical LULC changes affected extreme rainfall events.

1.2 Extreme rainfall during the 1993/2008 Midwest US floods

The Midwest floods of 1993 and 2008 were some of the costliest natural disasters in American history, totaling \$48 billion in damage and combining to 72 deaths (Lott et al., 2012). Warm season rainfall in the central United States is partly supported by the Great Plains low-level jet (GPLLJ) transporting water vapor from the Gulf of Mexico. The GPLLJ is a fast-moving low-level ribbon of air that can rapidly transport Gulf moisture and warmer temperatures to the north at speeds from 25-70+ knots. Other factors, such as positive land-atmosphere feedbacks and gradients in soil moisture anomalies, amplified rainfall in the 1993 floods. A positive feedback is the enhancement of an effect by its own influence, for example, precipitation leads to a more moist surface, which can increase the latent and sensible heat fluxes, increasing low-level energy available for convection.

The 1993 Midwest floods occurred from May 24, 1993, to June 23, 1993. This study will model from May 24, 1993 to June 13, 1993. This period had an abnormally high number of synoptic scale (~1000 km) troughs propagating across the central United States, allowing several Mesoscale Convective Systems (MCS) to occur. Peak flooding occurred in July and was associated with an anomalous trough in the western United States. Other contributing factors include increased moisture transport via the Great Plains Low-Level Jet (GPLLJ) and a strengthened and southward-shifted North American jet stream (Patricola et al., 2015). Over 10 mm/day of precipitation above average was recorded in portions of the western Midwest in July 1993. From onset through the peak floods, this period saw only roughly four days without more

than 0.05 inches of precipitation in our area of interest in or near our domain. Figure 1.1 and Figure 1.2 show the observed monthly precipitation across the contiguous United States (CONUS) for April, May, June, and July 1993 and 2008 respectively. Severe flooding occurred in the central United States and peaked in July with nearly 20 mm/day in areas along the Missouri and Iowa border.

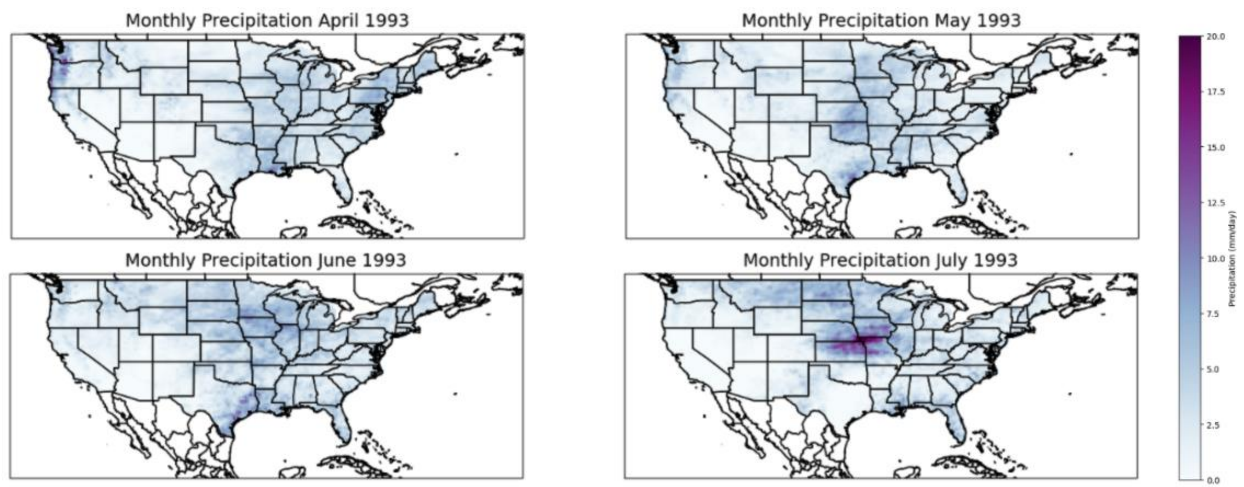


Figure 1.1. Observed monthly precipitation (mm/day) from CPC Unified Gauge-Based Analysis of Daily Precipitation over CONUS provided by NOAA PSL for the months of April, May, June, and July 1993. In what follows, the months of May, June, and July were simulated in the 50-km simulations, with the peak flood being in July.

The 2008 Midwest floods occurred from April 24, 2008, to June 13, 2008. Similar to the 1993 floods, the peak flooding was associated with an anomalous trough in the western US, a strengthened and southward-shifted North American jet, and increased moisture transport. Patricola et al., (2015) defines the peak of the 2008 floods as May 20, 2008, to June 13, 2008. From onset through the peak, there were only roughly ten total days without precipitation in our

area of interest. Severe flooding peaked in June across the central United States, with over 10 mm/day in some areas.

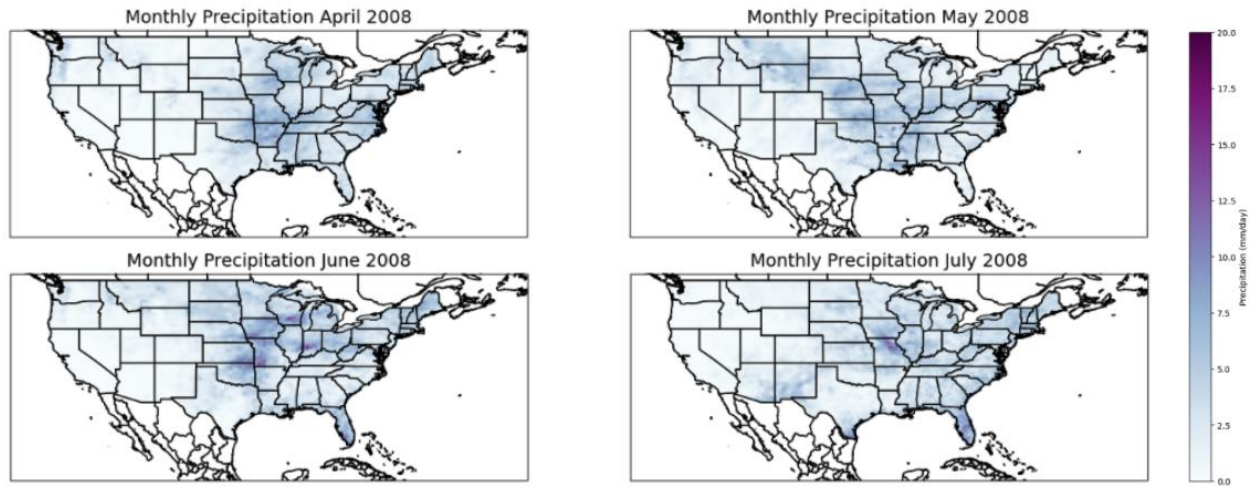


Figure 1.2. Observed monthly precipitation (mm/day) from CPC Unified Gauge-Based Analysis of Daily Precipitation over CONUS provided by NOAA PSL for the months of April, May, June, and July 2008. The months of May, June, and July are simulated in the 50-km simulations, with the peak flood being in June.

Our analysis will include effects on the regional circulation patterns (winds and pressure gradients), surface friction, PBL depth, and the low-level jet. We hypothesize that these changes will be driven by a change in albedo, sensible heat flux, and latent heat flux from land-use change.

1.3 Mechanisms of MCS, including land surface interactions

The goal of this study is to investigate how LULC changes affect the atmosphere. Much is still unknown about LULC changes and their effects on the weather extremes. At the time of this study, it is unknown how extreme rainfall events interact with changes in the low-level jet and boundary layer due to LULC change. An increase in surface roughness is anticipated to increase mixing within the boundary layer. Many other variables besides surface roughness will affect the atmosphere, such as latent heat flux, sensible heat flux, and albedo. This study will investigate several MCS events during the 1993/2008 Midwest floods to explore how the events may have been impacted by LULC change, including the low-level jet and surface wind response to the LULC change. The low-level jet is maximized in the evening and is responsible for transporting warm, moist air and assisting wind shear. The low-level jet is often a vital part of tornado outbreaks and the upscale growth of nocturnal MCSs. We anticipate that the change in LULC will alter weather events like these.

It is unclear how large of a difference LULC change could have on extreme rainfall due to uncertainties in climate model simulations related to their coarse resolution and use of convective parameterization. Therefore, earlier modeled deforestation effects on the water cycle may be unrealistic. For example, there are significant inconsistencies in modeled responses of rainfall extremes to LULC change (Pitman et al., 2012). Some studies show decreases in the frequency of afternoon precipitation and ‘MCS-like’ features following deforestation (Chen and Dirmeyer 2017; Devanand et al., 2020).

Convection-allowing model simulations (CAMs) differ from the parameterized simulations due to their ability to resolve cold pools and dynamic interactions with the LLJ and wind shear. We hypothesize that parameterized results are trigger-dominated, whereas realistic

MCSs respond to surface fluxes by increasing rainfall amount (duration/intensity). This hypothesis is consistent with CAM experiments (Dai et al., 2021), in which increasing evapotranspiration increased the rainfall amount rather than altering the frequency of MCS events.

There are several reasons deforestation could change MCS. Heavy rainfall-producing MCS are triggered and maintained by several mechanisms linked to land use, including mechanisms related to triggering, maintenance of the MCS, and precipitation rates. Some mechanisms for MCSs that may be affected by land use change are instability, lift, shear, water vapor, and precipitation efficiency. Precipitable water is the depth of water in a column of the atmosphere if all the water in the column was precipitated as rain. A change in precipitable water may affect the rain rates. Triggering mechanisms include buoyant instability and upward vertical motion (lift), linked to land use through sensible and latent heat fluxes, friction-causing eddies, moisture transport, gust fronts, and cold pools. Processes that maintain MCSs include cold pools and vertical shear of the horizontal wind, which may be affected by LULC change by an increased roughness length in forests, affecting the friction at the surface. Deforestation could also change the environment upstream and affect the propagation characteristics (longevity) of MCSs. Rainfall amounts are determined by precipitation efficiency, water vapor (precipitable water), and condensation rates (updated by updraft speed). Forests may provide higher near-surface water vapor due to high evapotranspiration, an increase in advection of water vapor due to changes in the LLJ, or through possible impacts of LULC change on large-scale circulation. Precipitation efficiency may also change with LULC change due to changes in updraft strength, warm versus cold rain in the cloud, and evaporation of precipitation below the cloud base.

1.4 Simulation methodology for the 1993/2008 Midwest US floods

This study focused on the 1993 and 2008 floods, using the Weather Research and Forecasting (WRF) model version 4.3.3 (Skamarock et al., 2021) to simulate the periods selected above. To implement the LULC experiments representing 1850 and present-day land vegetation scenarios, we changed the land-use index in the WRF input data to reflect the land-use and land cover in 1850. The simulations were rerun using the 1850 vegetation input data. Finally, the results of the flood period simulations with 1850 and present-day vegetation were compared to find changes in the water cycle caused by the land use and land cover change.

Convective precipitation is parameterized in climate models. Rainfall responses to LULC change in these models depend on convective parameterizations. Convective Allowing Model Simulations (CAMS) will differ from the parameterized simulations due to their ability to resolve cold pools and their dynamic interactions with the LLJ and wind shear. Two questions are posed in this study:

1. How has LULC change affected rainfall systems in the United States
2. How do modeled rainfall responses to LULC change depend on MCS dynamics?

Our hypothesis is that responses in parameterized experiments are likely to be more trigger dominated. For example, we expect parameterized convection to be sensitive to changes in convective available temperature (CAPE) that may result from the effects of LULLC on the water vapor content and temperature near the surface. CAPE is a measure of the amount of energy available for convection. Convective inhibition (CIN) is a measure of the amount of energy needed in order to initiate convection.

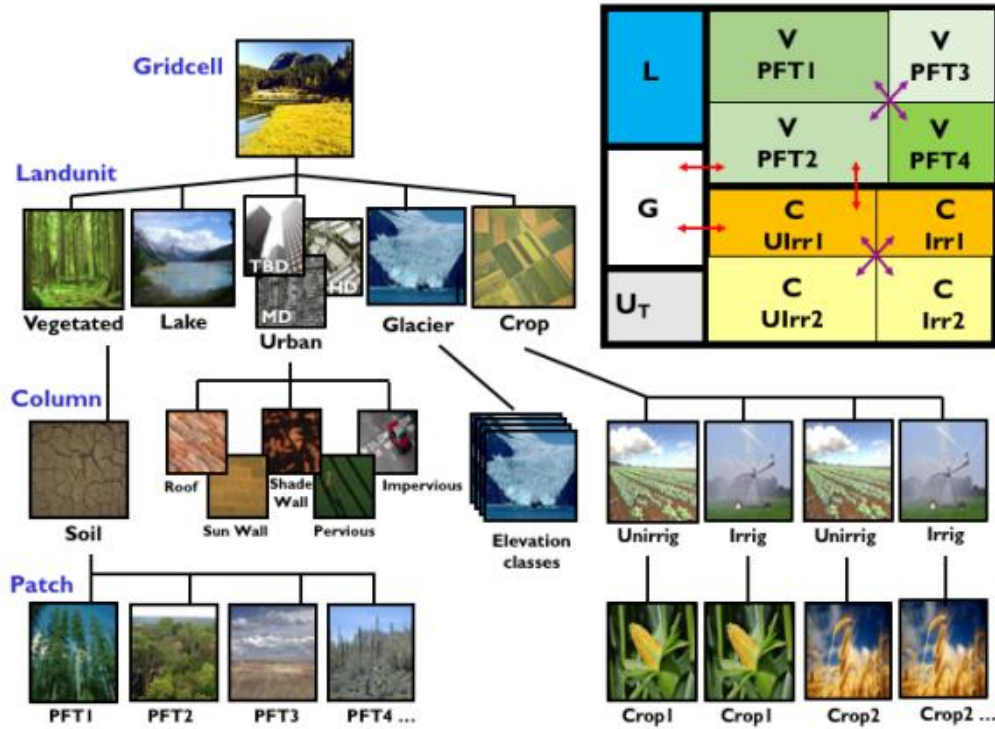


Figure 1.3. The CLM5 subgrid hierarchy. The upper right grid shows an example distribution of land units: lake (L), glacier (G), urban (U), different vegetation (V), plant functional types (PFTs), and crop (C) with rainfed (Rnfd) and irrigated crops (Irr). Figure from Lawrence et al., (2019a).

Historical land-use data for this study is from Community Earth System Model (CESM) runs performed as part of the Land Use Model Intercomparison Project (LUMIP). LUMIP provides gridded land-use data from a variety of modeling centers providing their estimates of land-use for scenarios in the past and future. The historical land-use scenario (land-hist) is based on the Land Use Harmonisation (LUH2 v1.0h) land use forcing data (850-2100). The land-hist experiment from the NCAR model CESM (CLM5 land model; land: CLM5 0.9x1.25) is used for our study. The protocol for their experiment is initialization in 1850 and allows spin up for 20

years with constant forcing. It should be noted that the CESM data represents a model prediction based on LUH land use forcing.

Using the CESM Plant Functional Type (PFT) and the International Geosphere–Biosphere Programme (IGBP) Modis Land-Use criteria, we translated the land use type from the CESM historical dataset (in PFTs) to the IGBP classification used in the Noah-MP Land Use Index (Table 1.1). This algorithm was tested on the present day and compared to Noah-MP to ensure the algorithm is reasonable. Then, the 1850 land use land cover dataset was created using the CESM PFT. The CESM PFT was spatially interpolated to the WRF grid. Since CESM allows multiple PFTs per grid cell, but Noah-MP used one, we used the CESM PFT percentage data to find the dominant plant type per grid cell based on the criteria in Table 1.1., and assigned a Land use index for Noah-MP. The present-day land use data was taken from the original WRF Noah-MP land use dataset. The WRF land use dataset was chosen rather than the CESM present-day dataset because it is based on observed vegetation whereas the CESM vegetation is predicted.

CESM Plant Functional Type (PFT)	→ Noah-MP Land Use Index
Tree Cover greater than 60% and dominated by Broadleaf Deciduous	Broadleaf Deciduous
Tree Cover greater than 60% and dominated by Broadleaf Evergreen	Broadleaf Evergreen
Tree Cover greater than 60% and dominated by Needleleaf Deciduous	Needleleaf Deciduous
Tree Cover greater than 60% and dominated by Needleleaf Evergreen	Needleleaf Evergreen
Tree Fraction greater than 60%. Dominated by neither deciduous nor evergreen (40-60% of either) tree type	Mixed Forest
Tree Fraction (10-30%)	Savanna
Tree Fraction (30-60%)	Woody Savanna
Wetlands dominate the land use in CESM dataset	Permanent Wetland
C3 or C4 grass dominate the land use in CESM dataset	Grassland
Crop Fraction dominates the land use in CESM dataset	Cropland
Cropland 40-60% and trees 10-60%	Cultivated/Natural Mosaic
Bare soil* plus shrub is greater than 60% and shrub fraction greater than 10% and less than 60%	Open Shrubland
Bare soil* plus shrub is less than 60%	Closed Shrubland

Table 1.1. CESM Plant Functional Type translation to Noah-MP Land Use Index.

Once the 1850 land use dataset was created using Table 1.1, the present-day land-use dataset was linearly smoothed to match the resolution of the CESM 1850 land use dataset. This avoided sharp discontinuities due to the resolution differences between input forcing and model grids. This study also developed high-resolution (3 and 15 km) land surface datasets that will serve as surface boundary conditions for future convection-allowing regional climate models (CAM) at 3-km resolution. In future work, it will be important to resolve updrafts, downdrafts, and cold-pools, to include realistic processes by which MCSs could respond to changes in US

forest cover. For example, forest roughness can reduce wind shear and deepen the cold pool (Dai and Williams 2022), while evapotranspiration can strengthen cold-pools (Schlemmer and Hohenegger 2014; Dai et al., 2020). Effects of evapotranspiration on convective triggering are also known to be resolution dependent (Hohenegger et al., 2009). Again, the present-day input datasets were smoothed because the WRF present-day input data was much higher resolution than CESM's 1850 input data. This resulted in 3-km and 15-km datasets of approximately the same resolution as the coarser CESM data for comparison (Figure 1.4). The same was done with the 50-km data to ensure the 1850 historical and present-day vegetation datasets have similar resolution (Figure 1.5). The 3-km and 15-km resolution datasets were prepared and saved, but not used for this study. Figure 1.5 shows the LUMIP Land-hist CESM Land Use from 1850 from the algorithm using the IGBP MODIS criteria and the Land Use dataset for the 50-km present day in the WRF Noah-MP model.

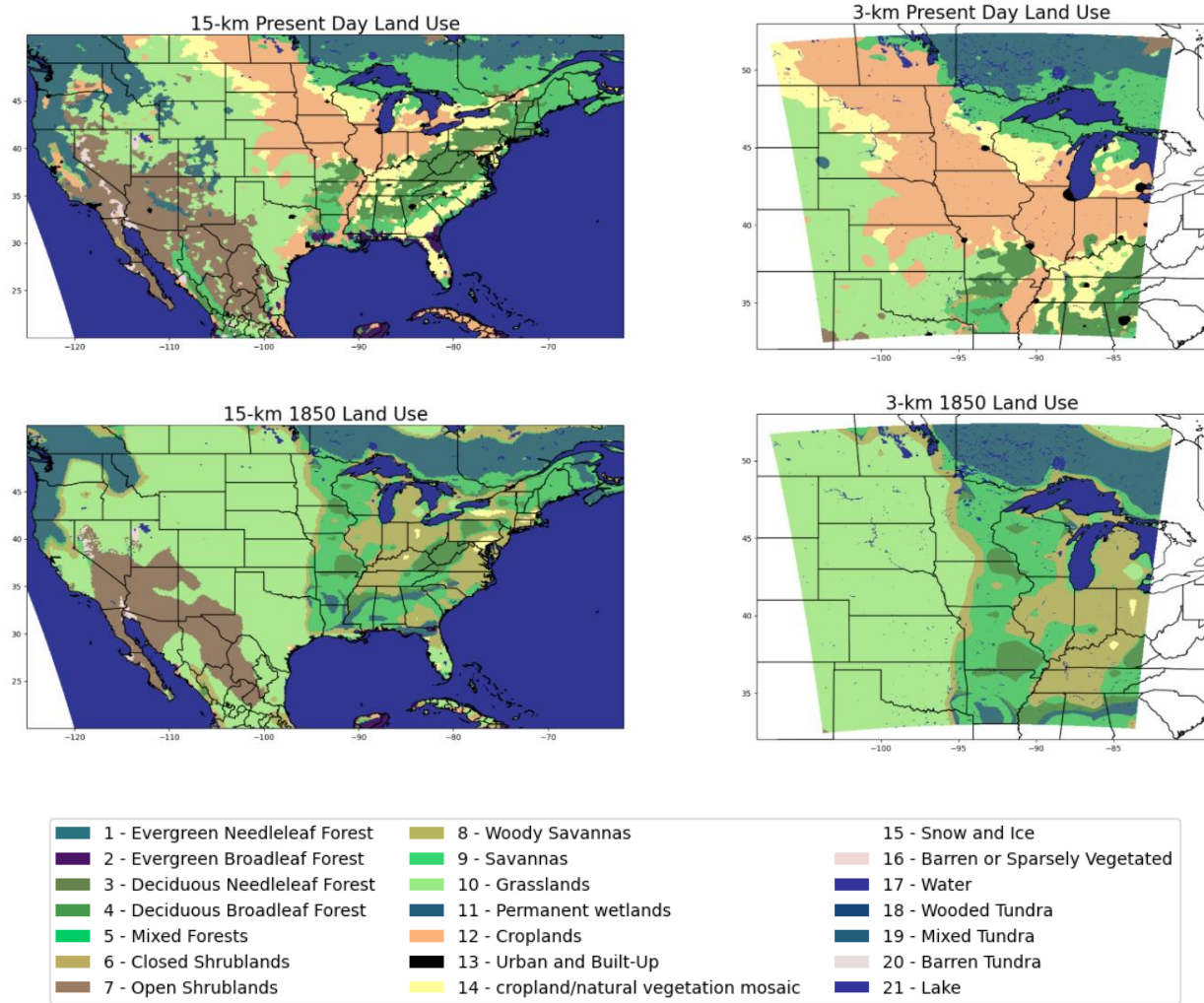


Figure 1.4. The present-day and 1850 land use index for the 15-km (left) and 3-km simulations (right). Both datasets were smoothed to reflect the coarser resolution of the historical LUMIP data.

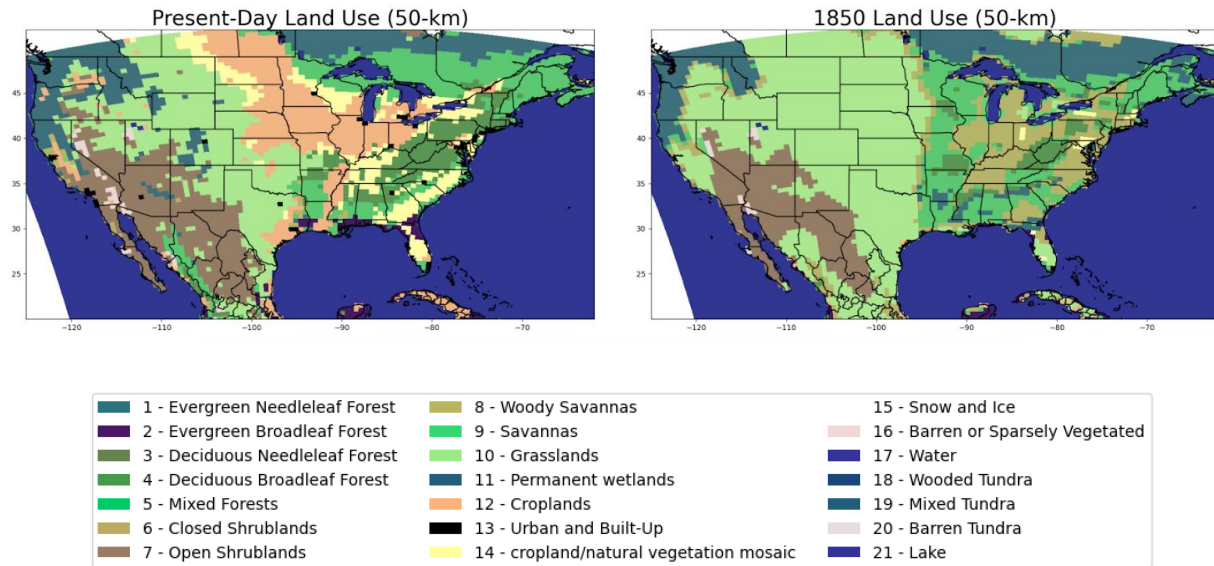


Figure 1.5. The present-day (left) and 1850 (right) land use index for the 50-km simulations. Both datasets were smoothed to reflect the coarser resolution of the historical LUMIP data.

The simulations for this study were run using the Weather Research and Forecasting (WRF) model version 4.3.3. The domain (Figure 1.6) is large enough to encapsulate the CONUS and much of the Atlantic Ocean, including the Bermuda High. It is also important to include the Gulf of Mexico because it provides a supply of water vapor that is converted to precipitation in Mesoscale Convective Systems.

WPS Domain Configuration

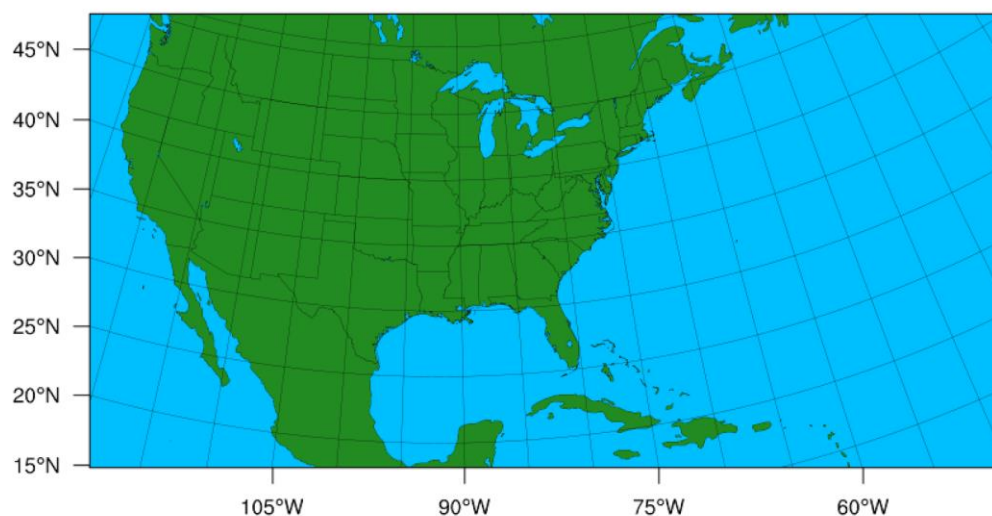


Figure 1.6. WRF Domain for the 50-km simulations. The domain captures the entire CONUS and the Bermuda High.

WPS Domain Configuration

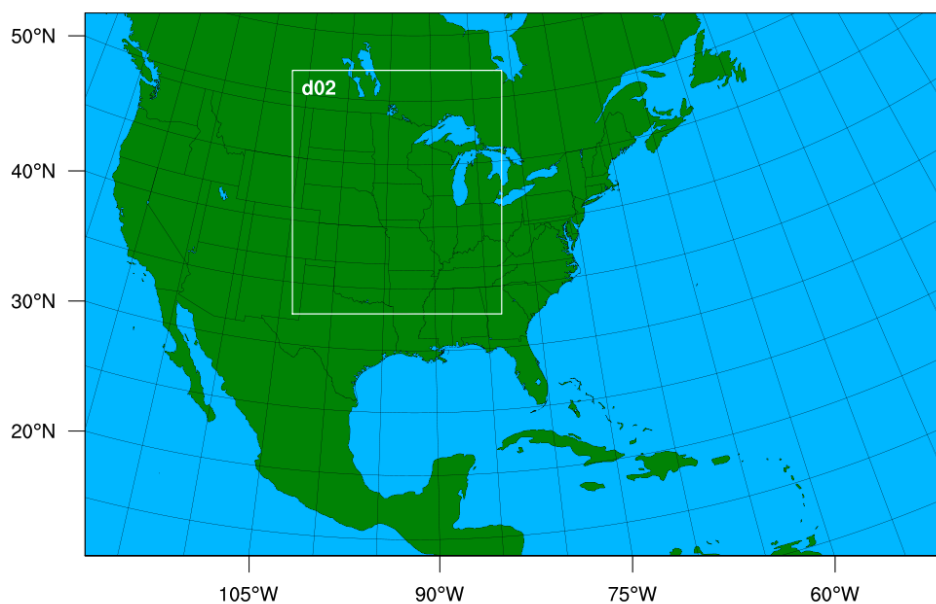


Figure 1.7. WRF Domain for the 3km nested simulations. The resolution is 15km in the parent WRF domain, and 3km in the nested domain (d02).

The simulations were run with 45 vertical levels and a 100 mb model top. Simulations will be run at 50-km (Figure 1.6) for the 1850 land use and the present-day land use and land cover. The surface and lateral boundary condition data used in the model simulations was from the European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 reanalysis data. The 3-km simulation has a parent domain of 15-km horizontal resolution and a 3-km nested domain (Figure 6). The 3-km nested domain is positioned so that all MCSs associated with the 1993 and 2008 floods develop within the domain. The 50-km and 15-km domain must have the convective parameterization turned on. The Betts-Miller-Janjic (BMJ) convective scheme was used. The convective scheme can be shut off for the 3-km nested domain. See chapter 2.2 for additional information on model parameterizations.

The radiation timestep is recommended to be 1 minute of simulation time per kilometer of grid spacing. For our 50-km runs, we chose a radiation timestep of 50 minutes between radiation physics calls. This value must be the same for all nested grids, so the 15-km and 3-km simulations have a radiation timestep of 18.

Resolution	Microphysics Scheme	Convective Scheme	Longwave Radiation	Shortwave Radiation	Boundary Layer	Surface Layer Physics	Surface Physics	Soil Levels	Surface Urban Physics	Vertical Levels
50 km	Thompson Scheme	BMJ	RRTM Scheme	RRTMG Scheme	MYNN 2.5 Level TKE Scheme	Monin-Obukhov (Janjic scheme)	Noah-MP Land Surface Model	4	None (Noah-MP)	45
3 km	Thompson Scheme	Off	RRTM Scheme	RRTMG Scheme	MYNN 2.5 Level TKE Scheme	Monin-Obukhov (Janjic scheme)	Noah-MP Land Surface Model	4	None (Noah-MP)	45

Table 1.2. WRF Model Parameterizations for the 50 km and CAM (3km) simulations

1.5 Land-use and land-cover in the Noah-MP model

The land-use land cover variables are changed in the WRF input files to get the desired configuration for our simulations. The land-use index is prescribed in the Noah-MP land surface model in each grid cell as one of 21 land-use types from the International Geosphere–Biosphere Programme (IGBP) vegetation classification scheme. The categories are listed in Figure 1.5.

Fractional vegetation (FVEG) is defined in Noah-MP as the greenness vegetation fraction. The green vegetation fraction is the fraction of a grid cell covered in green vegetation.

By definition, the leaf area index is the one-sided leaf area ratio per unit of ground area. LAI and stem area index (SAI) are in the MP table file (MPTABLE.TBL). With the Noah-MP namelist setting dynamic vegetation to option 1, dynamic vegetation is off; the model pulls LAI from MPTABLE.TBL, and prescribes FVEG from vegetation fraction (VEGFRA) in the WRF input files. The VEGFRA variable is also changed in the ‘WRF low input’ files (lower boundary condition files) to the desired FVEG dataset. This ensures that the LAI and FVEG used in the model correspond to the land-use index for the 1850 and present-day vegetation experiments. Noah-MP uses the table to assign LAI in the model based on the land use index. The change in LAI therefore reflects the change in land-use index (Figure 1.6). Note that we did not simply use LAI from the CESM model due to large differences in model physical parameterizations (subgrid tiles) and land use definitions that do not warrant a simple translation of LAI from one land model to another.

The albedo and heat fluxes are a result of very small-scale processes. Each leaf has pores called stomata in which water vapor and carbon dioxide are exchanged with the atmosphere. These processes are represented in the Noah-MP land surface model. Noah-MP scales up these processes to the grid cell level based on many variables. The most critical variables in our model

are albedo, vegetation fraction, and leaf area index. Albedo is a macroscopic consequence of the reflectance of shortwave radiation from leaves. A lower albedo allows the surface to absorb more shortwave radiation, (typically) increasing sensible heat flux. The vegetation fraction is used to scale the plant canopy processes (fluxes) up to the grid cell scale. WRF Noah-MP uses vegetation fraction to calculate latent and sensible heat flux.

A higher vegetation fraction means that a larger area of a grid cell is covered by vegetation. This affects the albedo and latent and sensible heat fluxes. LAI assigns the total leaf area over a ground of unit area and is used to scale these small-scale processes up to the grid cell together with vegetation fraction (Cowling and Field 2003). The Modified IGBP MODIS Noah vegetation dataset found in the MPTABLE assigns the LAI in Noah-MP.TBL file in WRF, and varies by LAI month also found in the MPTABLE.TBL file.

The WRF Noah-MP land surface model uses FVEG and LAI to calculate transpiration. First, to prepare for latent heat flux, Noah-MP calculates the water vapor conductance from leaf to canopy air (CTW) based on the dry leaf fraction. It then sums the conductance over the leaf area. Conductance from leaf to canopy air (CTW) (eq. 1) is calculated where FWET is the wetted fraction of the canopy, LAISUNE is the sunlit leaf area index, RB is the leaf boundary layer resistance, RSSUN is the sunlit leaf stomatal resistance, LAISHAE is the shaded leaf area index, and RSSHA is the shaded leaf stomatal resistance. CTW is the conductance of the dry leaves. If the leaves are wet, then the leaves evaporate like a pure water surface.

$$CTW = (1 - FWET) * (LAISUNE / (RB + RS_{SUN}) + LAISHAE / (RB + RS_{SHA})) \text{ (Eqn. 1)}$$

Next, Noah-MP calculates the transpiration flux (TR) by multiplying the conductance by the fractional vegetation, and then by the leaf-air vapor pressure gradient (ESTV-EAH). There are several thermodynamic factors like the psychrometric constant (GAMMAV) and air density (RHOAIR).

$$TR = FVEG * RHOAIR * CPAIR * CTW * (ESTV - EAH) / GAMMAV \text{ (Eqn. 2)}$$

where: CPAIR = 1004.64 (J/kg/K)

Since FVEG multiplies CTW, which is in turn a function of LAI, the most important factor for our LULC experiments is the product of the FVEG and LAI. One of the challenges of designing experiments with WRF-Noah MP is that there are four different places within input files and tables where vegetation fraction could be changed, which are not documented in the technical notes. Trial and error revealed that the VEGFRA variable within the wrflowinp and wrfinp files combined with Noah-MP namelist dynamic vegetation option “1” gives us the desired vegetation fraction and no dynamic vegetation. As the Noah-MP namelist suggests, LAI is taken from the table “MPTABLE.TBL” and prescribes LAI based on the land-use index (LU_INDEX). LAI has a different value for each month based on the LU_INDEX land use type. The FVEG from VEGFRA does not change with time as dynamic vegetation is off in our model. The effective leaf area index (ELAI) is the product of LAI and FVEG. The ELAI changes throughout the simulations due to the monthly variation in LAI (Figure 1.8, 1.9). The most notable changes between 1850 and present-day vegetation are where croplands replaced savannas and forests

from 1850 across the Midwest. The timing of our simulations should be noted when referring to the monthly ELAI where the 2008 case runs from April 24 to June 13, and the 1993 case runs from May 24 to July 13.

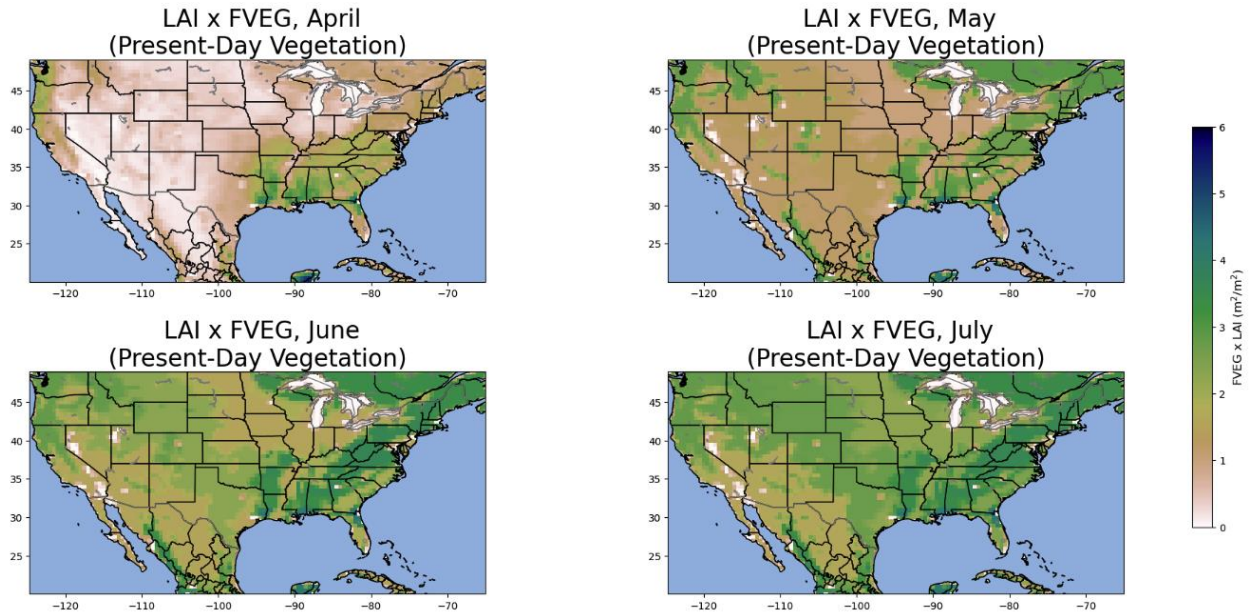


Figure 1.8. Present-day leaf area index (LAI) multiplied by vegetation fraction (FVEG) for the months of April, May, June, and July.

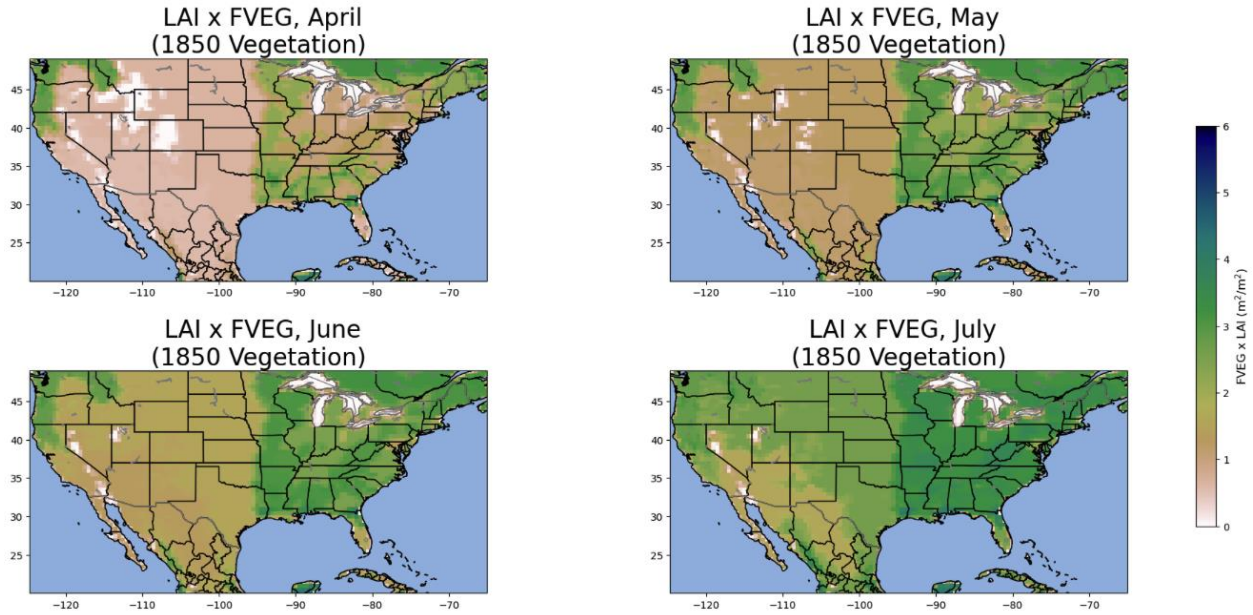


Figure 1.9. 1850 leaf area index (LAI) multiplied by vegetation fraction (FVEG) for the months of April, May, June, and July.

Changes in vegetation can affect the surface wind speeds due to the change in roughness length. The roughness length does not exactly correspond to any physical length. A larger roughness length increases friction at the canopy layer (which acts as the surface displaced a distance, d) and increases eddy flux. The ‘extra’ induced turbulence of a rough surface will decrease wind speeds close to the surface (Stull 1988). Roughness length increases as a surface becomes less smooth. The sea has a roughness length of near 0, prairies or farm fields increase roughness length to 0.3, and forest clumps or scattered buildings can increase roughness length to 0.25-0.5 (The Davenport Classification).

CHAPTER 2. IMPACTS OF US DEFORESTATION ON RAINFALL FROM MESOSCALE CONVECTIVE SYSTEMS

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2.1 Introduction

Extreme weather events have been attributed to global warming and rising greenhouse gas concentrations. However, regional climate change has also resulted from extensive clearing of US forests for agriculture since the 1800s (Bonan, 1999) and from the intensification of agricultural productivity since the 1950s (Alter et al., 2018). Recent efforts toward reforestation could further alter regional and global climate (Lawrence et al., 2022). Forests affect the radiative properties (albedo) of Earth's surface (Rohatyn et al., 2022; Bonan, 1999), its aerodynamic roughness (Eder et al., 2015; Bonan et al., 2018), and the two-way exchanges of water in the forms of evapotranspiration and canopy interception of precipitation. In particular, the uptake of atmospheric CO₂ for photosynthesis necessitates an evaporative loss of water in the form of transpiration. Globally, transpiration is the largest component of evapotranspiration (Jasechko et al., 2013). The coupling between photosynthesis and transpiration could intensify the water cycle in reforested areas (Odoulami et al., 2019), depending on how much water returns as rainfall.

An increase in rainfall in response to increased soil moisture would represent a positive feedback. Positive soil moisture-rainfall feedback is thought to have contributed to the extreme

rainfall during the 1993 Midwest US floods (Trenberth and Guillemot 1996; Beljaars et al., 1996), and to rainfall deficits during the 2012 US drought (Saini et al., 2016). However, there is disagreement among modeling studies on the response of rainfall to changes in forest cover (Pitman et al., 2012). Based on climate model experiments, preindustrial deforestation in the US resulted in significant regional cooling (Bonan 1999) and a decrease in the frequency of afternoon convective precipitation (Chen and Dirmeyer 2017, Devanand et al., 2020), with little or no effect on average precipitation (Bonan, 1999; Trail et al., 2015). These findings are subject to uncertainty due to convective parameterizations that employ a threshold value of convective available potential energy (CAPE) to trigger convection.

While extreme rainfall can be linked to anomalous continental-scale atmospheric circulations, heavy rain is typically produced by smaller-scale thunderstorm clusters (Fritsch et al., 1986), known as mesoscale convective systems (MCSs). MCSs are not adequately resolved in conventional climate models (Prein et al., 2020). Moreover, soil moisture-rainfall feedbacks depend on both the environmental state (Findell and Eltahir 2003a,b; Williams 2019; Findell et al., 2011) and cloud-scale processes (Hohenegger et al., 2009), highlighting the importance of resolving thunderstorms updrafts and downdrafts at kilometer scales. Rain-cooled air masses underneath MCSs, known as cold pools, aid in triggering convection and maintaining the organized updrafts and downdrafts within MCSs (Rotunno et al., 1988).

Biases often occur in coarse resolution models. A summertime warm/dry bias occurs in climate models. For example, an overestimate of surface air temperature by up to 6°C occurs in the National Oceanic and Atmospheric Administration GFDL climate model AM2 within the first few days of the simulation, and this warm bias can persist due to insufficient precipitation (Klein et al., 2006). Warm biases are also present in other models with convective

parameterization, such as the NCAR Community Earth System Model (CESM). The summer temperatures in CAM4 and CAM5 are too high when compared to observations and reanalysis. The amplitude of the diurnal cycles are also overestimated in CAM4 and CAM5 (Lindvall et al., 2013). Other studies investigate the warm biases in the Atmospheric Model Intercomparison Project (AMIP). This study found a land-atmosphere interaction hotspot region in the central United States while investigating biases in simulation summer surface air temperatures (Merrifield et al., 2016). Another study found surface temperature biases in the southern Great Plains are associated with large shortwave and longwave biases in climate and numerical weather prediction models (Weverberg et al., 2018). Warm biases are well documented in coarse resolution models.

The development and propagation of MCS is potentially influenced by land-use and land cover. Some mechanisms for MCSs that may be affected by land-use change are instability, lift, shear, and precipitation efficiency. Instability is linked to land use through sensible and latent heat fluxes, lift can ultimately be linked to land use through increased surface friction causing eddies. Shear is affected by LULC change by an increased roughness length in forests affecting the friction at the surface, thus changing the wind shear. Precipitation efficiency can also change with LULC change due to updraft strength, warm versus cold rain in the cloud, and evaporation of precipitation below the cloud base.

Our research question for this study is how deforestation might impact rainfall from MCSs. Mechanisms for MCSs, such as moisture convergence, maximum low-level winds, and instability, can all be affected by land use land cover change. Convective available potential temperature (CAPE) is a measure of the amount of energy available for convection. Convective inhibition (CIN) is a measure of the amount of energy needed in order to initiate convection.

Deforestation could change the environment upstream, including the convective available potential energy and convective inhibition energy. Most notably, changes in momentum transfer, heat fluxes, and shear are expected with LULC change. Large scale circulations may experience changes due to changes in sensible heat fluxes, possibly changing the pressure gradients and low level jet magnitude. Changes in surface friction could affect the longevity and strength of MCSs by changing the characteristics of cold pools (depth and propagation speed). There could also be changes to large scale circulations due to changes in heat fluxes which can change the environment for MCS, including the strength and location of the low-level jet, and moisture convergence.

Our experiment will investigate the changes in precipitation in the United States for the 1993 and 2008 flood events. Other studies have looked at the overall average precipitation not necessarily connected to MCSs and flood events. We are interested in the precipitation maxima during these events as well as the event average rainfall at times of flooding. Other studies have investigated the changes in average precipitation for land use land cover change, and found a decrease in overall precipitation (Chen et al., 2017).

The 1993 flood event has been described as a series of quasi-stationary convective events. A quasi-stationary MCS led to considerable flooding across the Midwest, particularly Iowa. The so-called ‘MBE’ wind vectors are a forecasting tool to diagnose storm motion (Junker and Schneider, 1997). The MBE wind vectors suggest many MCSs during this flood event, particularly in our July 8, 1993 case study, were moving at very slow speeds, allowing severe flooding to occur. Other studies on this flood event found that significant convection developed above a stable boundary layer, meaning many MCSs during this period were elevated (Rochette and Moore, 1996). Their study also found a low level jet maximum oriented normal to the

surface boundary, and high levels of surface and mid-level moisture were some of the important factors needed for these elevated extreme precipitation events. The 2008 flood event is similar to the 1993 flood event where there was an anomalous number of synoptic troughs, a strengthened and southward shifted low level jet, and increased moisture transport.

2.2 Methods

The potential area of reforestation in the contiguous United States is estimated to be over 100 million acres (Cook-Patton et al., 2020), or roughly half of the forested area lost since European colonization (Ramankutty, Heller, and Rhemtulla 2010). As an upper bound on potential forest cover in the US, we prescribed land-use and land-cover from the NCAR Community Earth System Model (CESM2) at $0.9^\circ \times 1.25^\circ$ resolution, using the preindustrial (year 1850) scenario of the land-use model intercomparison project (LUMIP) (Lawrence et al., 2016). We ran two experiments using the Weather Research and Forecasting (WRF) model to compare 1850 land-use and land-cover (LULC) to present-day LULC. The model domain included the Gulf of Mexico and western Atlantic basin to allow the simulated pressure gradients enough distance to realistically adjust to land-surface changes (Seth and Giorgi, 1998).

The Weather Research and Forecasting (WRF) model version 4.3.3 was used to simulate the historical flooding events. The models will be run at 50 km resolution similar to most climate models. Each model run simulates the 1993 floods and the 2008 floods with present-day land use and 1850 land use. The domain is large enough to encompass the CONUS and Bermuda High (Figure 2.1).

WPS Domain Configuration

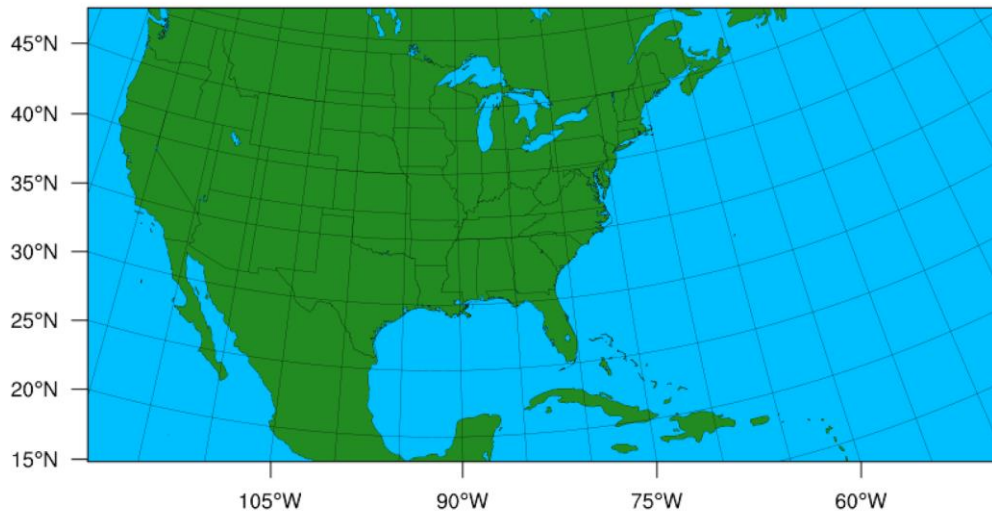


Figure 2.1. WRF domain for the 50-km simulations. The domain captures the entire CONUS and the Bermuda High.

We used the following parameterizations: Thompson microphysics, RRTM longwave radiation, RRTMG shortwave radiation, MYNN 2.5 Level TKE boundary layer, Monin-Obukhov surface layer physics, and the Noah-MP Land Surface Model, with four soil levels and no surface urban physics. Noah-MP includes urban physics independently; thus, we do not set a surface urban physics scheme (Table 1.2). The 50-km resolution model used the BMJ convective scheme. Dynamic vegetation option was “off” and vegetation LAI was assigned using the Noah-MP table LAI, which is based on land use type. This option also uses the maximum vegetation fraction from the WRF input files for the present day.

Historical land-use data for this study is from the Community Earth System Model (CESM) from the Land Use Model Intercomparison Project (LUMIP). LUMIP is a project aimed at modeling land use change as human land use expands. This data is a gridded land use dataset

providing data for the past, and future scenarios. The data is from Land Use Harmonisation (LUH2 v1.0h) land use forcing data (850-2100). The land-hist experiment from the NCAR model CESM (CLM5 land model; land: CLM5 0.9x1.25) is used for our study. The protocol for their experiment is to initialize in 1850 and allow a spin up for 20 years with constant forcing. The CESM data represents a model prediction based on LUH land use forcing. It should be noted that this historical land use data is a model prediction.

CESM Plant Functional Type (PFT)	→ Noah-MP Land Use Index
Tree Cover greater than 60% and dominated by Broadleaf Deciduous	Broadleaf Deciduous
Tree Cover greater than 60% and dominated by Broadleaf Evergreen	Broadleaf Evergreen
Tree Cover greater than 60% and dominated by Needleleaf Deciduous	Needleleaf Deciduous
Tree Cover greater than 60% and dominated by Needleleaf Evergreen	Needleleaf Evergreen
Tree Fraction greater than 60%. Dominated by neither deciduous nor evergreen (40-60% of either) tree type	Mixed Forest
Tree Fraction (10-30%)	Savanna
Tree Fraction (30-60%)	Woody Savanna
Wetlands dominate the land use in CESM dataset	Permanent Wetland
C3 or C4 grass dominate the land use in CESM dataset	Grassland
Crop Fraction dominates the land use in CESM dataset	Cropland
Cropland 40-60% and trees 10-60%	Cultivated/Natural Mosaic
Bare soil* plus shrub is greater than 60% and shrub fraction greater than 10% and less than 60%	Open Shrubland
Bare soil* plus shrub is less than 60%	Closed Shrubland

Table 2.1. The CESM Plant Functional Type (PFT) to Noah-MP Land Use Index conversion table used to prescribe historical land use in the simulations.

Using the CESM Plant Functional Type (PFT) and the IGBP Modis Land-Use criteria (Table 2.1), we defined land use type in the historical dataset and represented this data in the Noah-MP Land Use Index (Table 2.1). This algorithm was tested on the present-day vegetation and compared to Noah-MP to ensure the algorithm is reasonable. Then, the 1850 land use land cover dataset was created using the CESM PFT. The CESM PFT data must be interpolated to the WRF grid. Once this is done, we can use the CESM PFT to find the dominant plant type based on the criteria in Table 1., and assign a Land-Use Index that can be understood by Noah-MP.

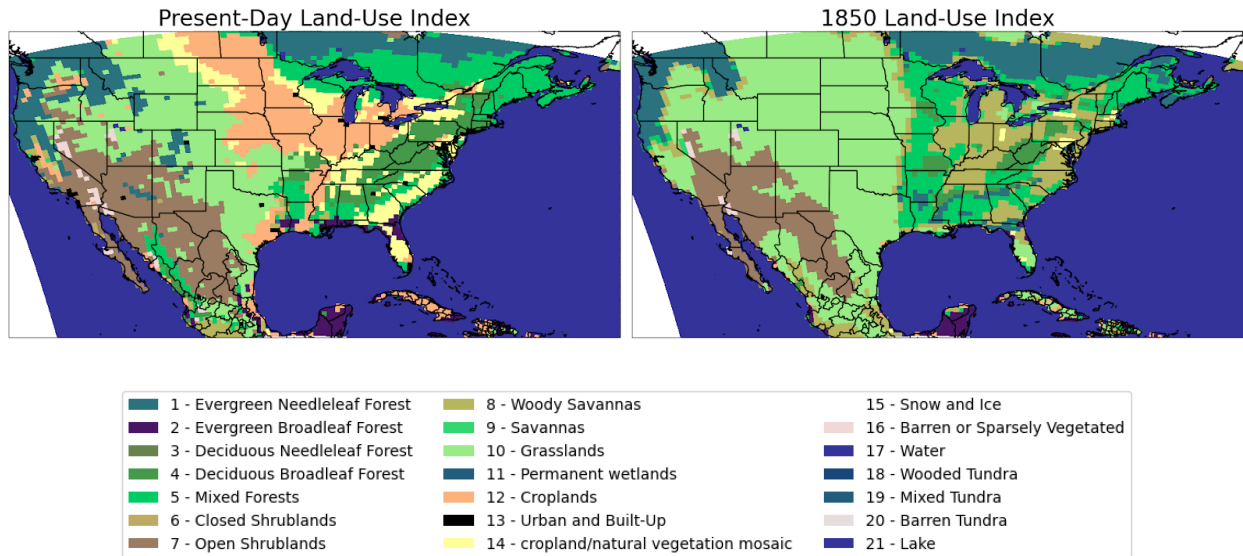


Figure 2.2. Noah-MP IGBP MODIS Land-Use Index for present-day (left) and 1850 (right) for the 1993 and 2008 50-km simulations. The present-day vegetation is from WRF Noah-MP, and the 1850 vegetation is from the CESM PFT data adjusted to the Noah-MP IGBP MODIS land-use type.

Rainfall observation data was taken from the CPC Unified Gauge-Based Analysis of Daily Precipitation over CONUS (Xie et al., 2007). This dataset provides daily accumulated precipitation, starting from 12:00 UTC the day prior to 12:00 UTC the day of, that was used to compare the model precipitation to the observed precipitation.

The roughness length was prescribed based on a land-use index in the Noah-MP land surface model using a table (Table 2.2). Roughness length increased by about 60% with 1850 vegetation from the Ozarks to the Upper Mississippi River Valley (Figure 2.3). This area corresponds to where the 1850 experiment prescribed forests in place of present-day crops. In other areas of the Southeast US, the 1850 experiment prescribed woody savannas in place of crops. Roughness length increased from about 0.05 with 1850 vegetation to 0.15 in the present-day where crops and other vegetation replaced the woody savannas in parts of Indiana, Kentucky, and Tennessee.

Vegetation Type	Roughness Length
Evergreen Needleleaf Forest	0.50
Evergreen Broadleaf Forest	0.50
Deciduous Needleleaf Forest	0.50
Deciduous Broadleaf Forest	0.50
Mixed Forest	0.20 - 0.50
Woody Savanna	0.01 - 0.05
Grasslands	0.10 - 0.12
Water	0.0001
Cropland	0.05 - 0.15

Table 2.2. Vegetation type and its corresponding maximum roughness length in the Noah-MP land surface model. Some vegetation types have a roughness length that varies by month. This table only includes the relevant land-use types in our model.

It is perhaps surprising that the 1850 roughness length decreased relative to the present-day over parts of the Ohio Valley and East Coast (North Carolina/Virginia). This reflects the shorter roughness lengths of savanna compared to crops (Table 2).

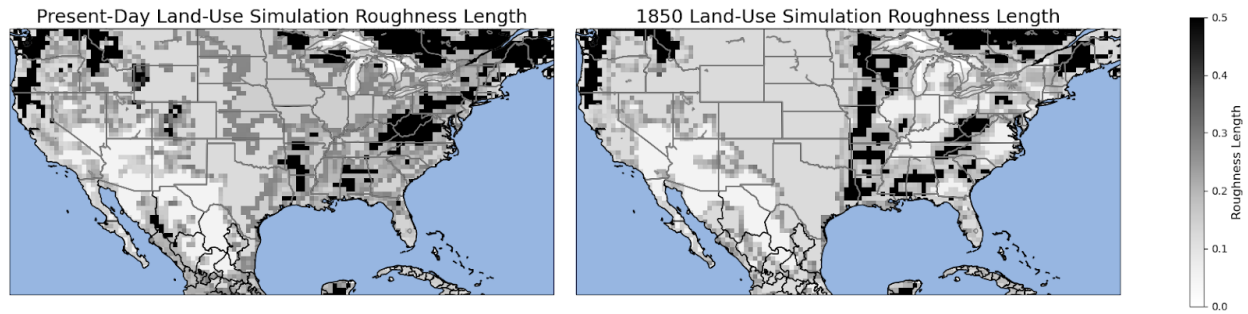


Figure 2.3. Present-Day (left) and 1850 (right) maximum vegetation roughness length values from the Noah MODIS IGBP vegetation parameters. Darker colors indicate a larger roughness length.

2.3 Results

2.3.1 1993 Flood Event

Over the 1993 study period (24 May to 13 July), the prescribed 1850 LULC had the effect of increasing vegetation amount over much of the US Midwest and Southeast (Figure 2.4). We used an effective leaf area index (ELAI) to quantify the combined effects of the fractional cover of each vegetation type (FVEG) and its leaf area index. ELAI is defined as the product of FVEG and LAI. ELAI where there are crops in the present-day vegetation simulation is around 1.0 (unitless) in May and increases to 2.5 in July (not shown in figures). ELAI in the 1850-vegetation simulations increased by 1-2 except for over the Appalachian region. The ELAI was greater in 1850 mostly due to forests existing where there is crop cover in the present-day.

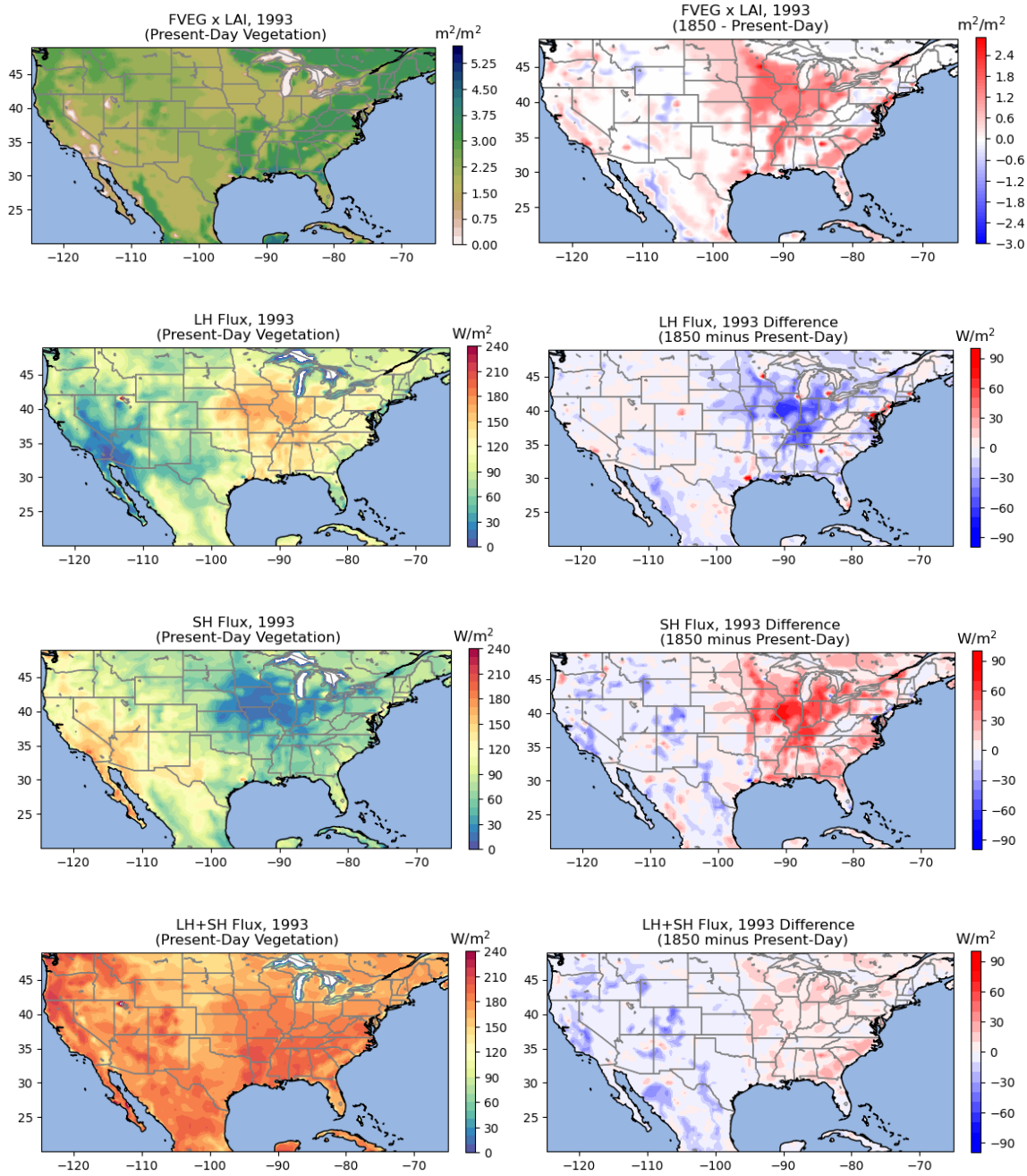


Figure 2.4. Present-day and 1850 vegetation experiments and corresponding changes in surface heat fluxes for the 1993 flood case. (a) Present-day vegetation effective LAI (FVEGxLAI). (b) Change in ELAI (1850 minus present-day vegetation). (c-d) As in a-b but for latent heat flux (e-f) As in a-b but for sensible heat flux. (g-h) As in (a-b) but for the total heat flux.

Latent heat flux decreased with 1850 vegetation (Figure 2.4c,d), in the same areas where ELAI increased. In other words, the change from crops to forests appears to have decreased the latent heat flux. This is consistent with the higher latent heat flux in the present-day experiment over parts of the agricultural Mississippi Valley and Midwest where modeled present-day ELAI is generally less than 2. Sensible heat flux increased from 0-50 W/m² to 70-130 W/m² with 1850 vegetation. Daily sensible heat in the present-day over parts of Missouri, Iowa, and Illinois averaged to nearly 0 W/m², implying a very high evapotranspiration rate over croplands in Noah-MP. Total heat flux with 1850 vegetation was 10-30 W/m² greater than present-day (Figure 2.4g,h), meaning that forests increased the energy available for turbulence (net radiation minus ground heat conductance) at the surface.

Albedo decreased by about 30-40% with 1850 vegetation (Figure 2.5a, b). A smaller albedo could explain the greater total heat flux with 1850 vegetation because more shortwave radiation is being absorbed at the surface. Albedo decreases in areas with the larger increases in ELAI (e.g. across the upper Midwest), which is consistent with a darker surface in forests compared to crops. Albedo does not explain the increase in temperature over the Northern Plains. There, we see that the sensible heat flux increased as latent heat flux decreased, suggesting that the temperature increase there is due to a shift in the partitioning of heat fluxes (less transpiring vegetation). Crops in the present-day replaced grasses in this region, suggesting that the grasses transpire less than the crops.

The 2-meter air temperature increased with the 1850 vegetation from present-day (Figure 2.5c,d). The 1850 vegetation 2-meter air temperature was over 4°C higher than the present-day surface temperature in some areas. In general, most areas had an increase in temperature anywhere from 1°C to 5°C with 1850 vegetation. This increase in 2-meter air temperature

reflects the increase in sensible heat flux over the eastern portion of the US, stemming from the change in ELAI from present-day to 1850 where the darker forests (lower albedo) result in greater sensible heat flux.

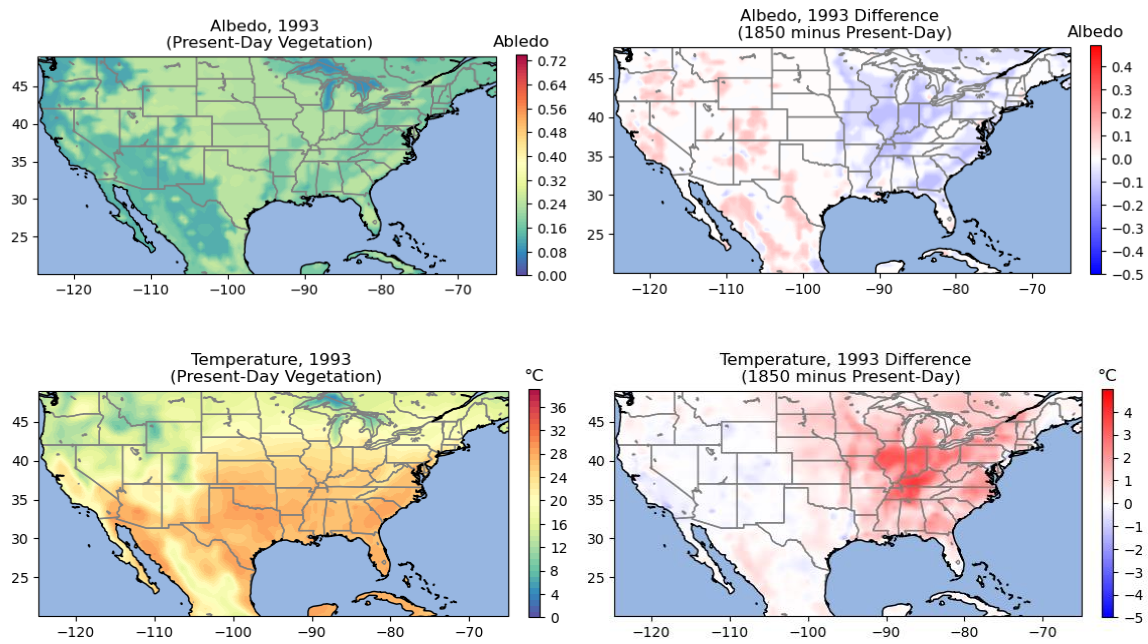


Figure 2.5. Average surface albedo change and 2m air temperature from the 1993 flood case simulations. (a) Average albedo in the present-day vegetation experiment (b) Albedo change (1850 minus present-day vegetation) (c) Present-day temperature (d) Temperature change (1850 minus present-day vegetation).

The average sea-level pressure decreased with 1850 vegetation across much of the eastern United States, by up to 1-3 mb (Figure 2.6a, b). The pressure dropped the most where there was an increase in ELAI in the 1850 vegetation. These areas broadly correspond to where the surface temperature increased in the 1850 vegetation experiment, suggesting a thermal

response. The decreased eastern US sea level pressure decreased the pressure gradient from the eastern US to the Rockies.

10-meter wind speeds decreased in the 1850-vegetation simulations where deforestation occurred for agriculture in the present-day (Figure 2.6c, d). This is likely due to the increased surface roughness with forests. An area from the Ozarks through the Mississippi River Valley had the largest decrease in wind speed of up to 50% with 1850 vegetation, corresponding to the region where 1850 forests were replaced with crops in the present-day. The decrease in wind speed can be attributed to an increase in roughness length in forested areas (as shown previously).

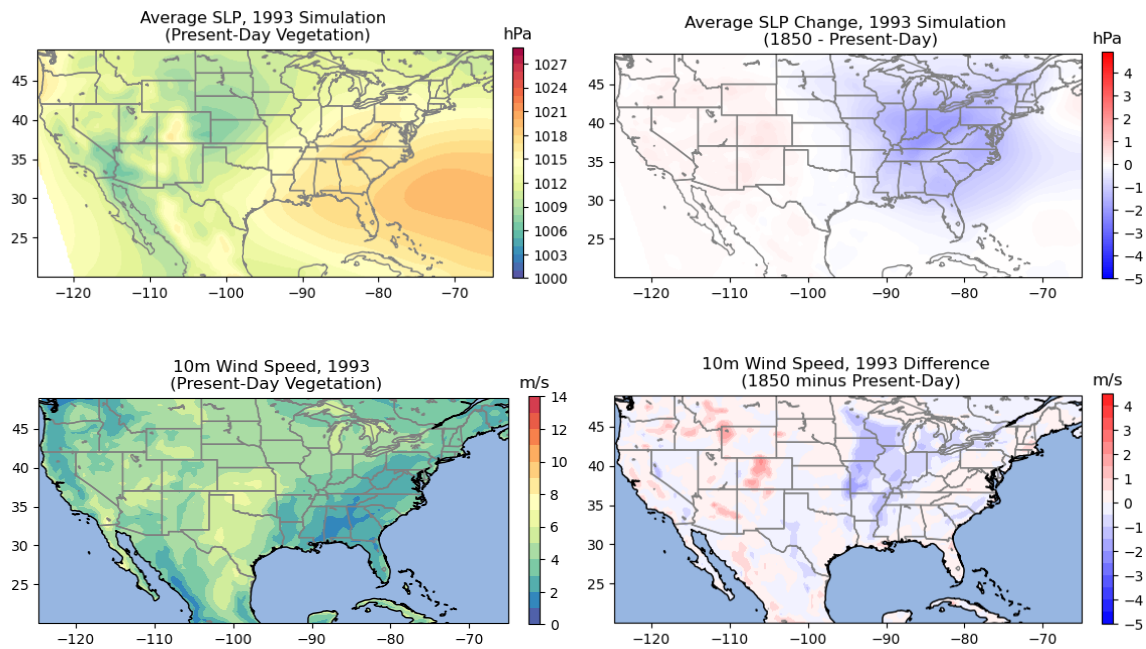


Figure 2.6. Average sea level pressure and 10m wind speed from the 1993 flood case simulations. (a) Present-day average sea level pressure (b) Sea level pressure change (1850 minus present-day) (c) Present-day average 10m wind speed (d) 10m wind speed change (1850 minus present-day).

Water vapor decreased in the 1850-vegetation simulations especially where there was a decrease in latent heat flux. 2-meter water vapor mixing ratio decreased by over 10% across much of the central and eastern United States (Figure 2.7a, b). Precipitable water also decreased significantly over the central and eastern US (Figure 2.7c, d). This decrease in precipitable water is likely due to decreased moisture advection from a weakened LLJ and decreased latent heat flux at the surface. Rochette and Moore (1996) found that high levels of surface moisture and precipitable water were crucial to these flooding events, and the decrease in these variables with the 1850-vegetation simulations would lead us to expect a decrease in precipitation.

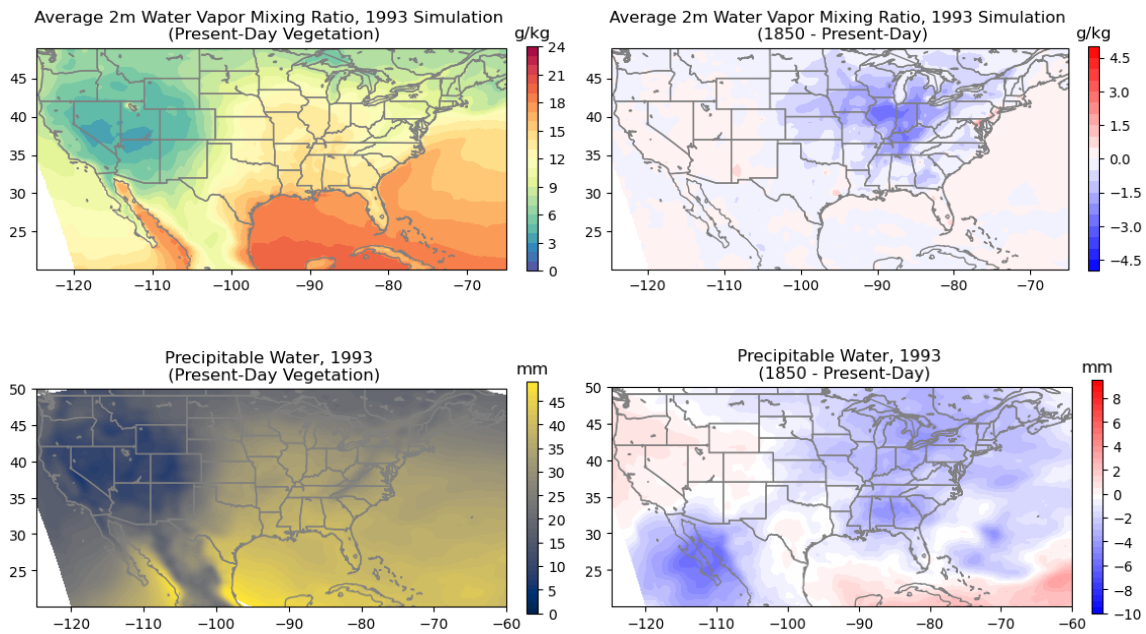


Figure 2.7. Average water content variables from the 1993 flood case simulations. (a) Present-day average 2m water vapor (b) 2m water vapor change (1850 minus present-day) (c) Present-day precipitable water (d) Precipitable water change (1850 minus present-day).

Winds at the low-level jet altitude (Figure 2.8a, b) were examined to understand the contribution of water vapor transport to the surface water vapor mixing ratio and precipitable water. The overall pattern of 850 mb wind shows the climatological low-level jet over the Plains and Midwest. The 850 mb wind speeds decreased by up to 10% over the Plains and Midwest in the 1850-vegetation simulations relative to the present-day. This could be due to a weakened LLJ, limiting the transport of warm, moist air.

To understand the changes in wind, we turn to the geopotential height on constant pressure surfaces (Figure 2.8e, f). The climatological distribution of modeled height shows a broad gradient of high to low heights from the US Southeast to Newfoundland, resembling the climatological Bermuda High and western extent of the Icelandic Low. Heights increased off the coast of Maine by 1-4 meters. The pressure gradient in the Atlantic (between the climatological high and low) decreased in the 1850-vegetation simulations. Winds increased in the area of weakening climatological high pressure. 850 mb pressure heights decreased 4-8 m in the east and southeast in the 1850-vegetation simulation, similarly to the decrease in sea level pressure shown earlier. The decreased 850 mb pressure height gradients and 850 mb wind speeds from the Gulf of Mexico with 1850 vegetation affects the mesoscale convective systems and rainfall rates. In the 1850-vegetation experiment there is a weakening of the western extension of the Bermuda High over the southeast US and Ohio Valley. This result can be explained by the increase in surface temperature in the 1850 vegetation experiment, which lowers the height of the 850 mb surface and reduces the pressure gradient force for large scale circulations. Around this weakening of the high, there is an increase in pressure level heights over the Plains and off the coast of Maine. In summary, the decreased pressure gradients across our domain led to a

weakened LLJ in the 1850 vegetation experiment, which can weaken the transport of moisture and warm air to the Plains and Midwest and affect MCS precipitation.

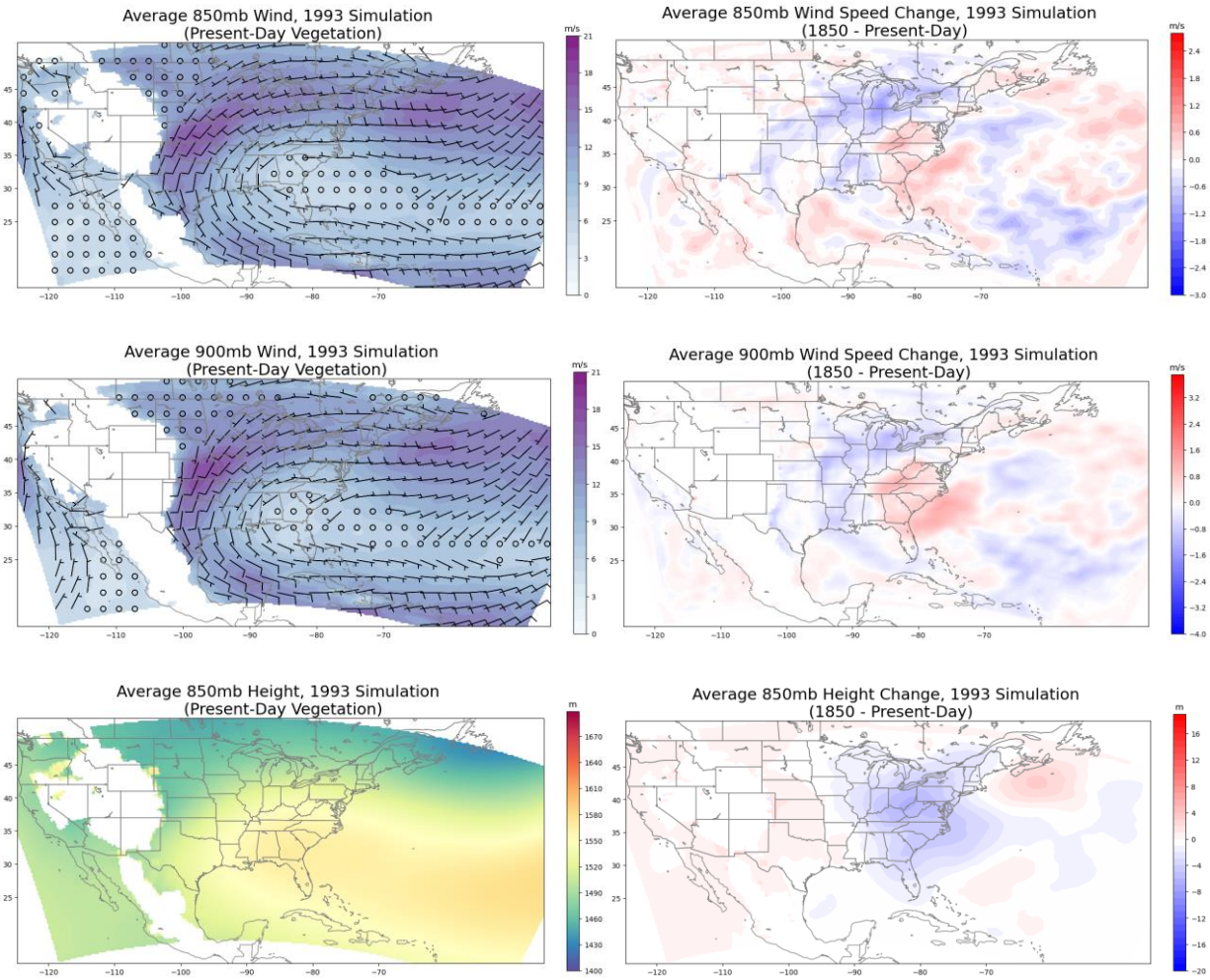


Figure 2.8. Responses of low-level wind speed and geopotential heights in the 1993 experiments. (a) Average 850 mb wind speed (m/s) and direction (m/s) with present-day vegetation, (b) 850 mb wind speed change (1850 minus present-day vegetation). (c-d) as in (a-b) but for 900 mb wind speed (m/s) and direction (m/s) (e-f) As in (a-b) but for 850 mb height. Wind barbs in (m/s).

The factors contributing to convective triggering include CAPE, CIN, and PBL height. The average mixed layer CAPE values over the Midwest decreased by up to 400 J/kg in the 1850-vegetation simulations (Figure 2.9a, b). The decrease in CAPE may be explained by the decrease in water vapor at the surface. An increase in surface temperature and a decrease in water vapor at the surface will raise the lifted condensation level (LCL), requiring air parcels to travel a greater distance before becoming saturated. This will decrease CAPE because it will take longer for air parcels to become saturated and buoyant in the atmosphere.

Convective inhibition (CIN) changes were more variable than other parameters from 1850 to present-day (Figure 2.9c, d). The increase in CIN in the Southeast US appears to correspond with the increase in sensible heat and PBL height in the 1850 vegetation experiment (as shown earlier). A deeper PBL allows for more mixing and entrainment, which could make it more difficult to initiate convection by drying the PBL. CIN decreased in the 1850-vegetation simulations over the southern and central Plains, although it does not appear to be a significant decrease. An increasing PBL height is usually associated with an increase in sensible heat flux. The increase in sensible heat flux warms the boundary layer and increases the temperature at the LCL, which could also work to decrease CIN. These competing factors can make it generally difficult to diagnose variations in CIN.

PBL heights increased by nearly 40-50% with 1850 vegetation (Figure 2.9e, f). The average PBL height with 1850 vegetation was 400 meters higher than in the present-day where present-day crops changed to forests and woody savannas in the 1850 experiment, again indicating that forests led to greater sensible heat flux and less latent heat flux in Noah-MP. A higher sensible heat flux contributes to increased buoyancy of dry air parcels in the boundary layer, which deepens the PBL.

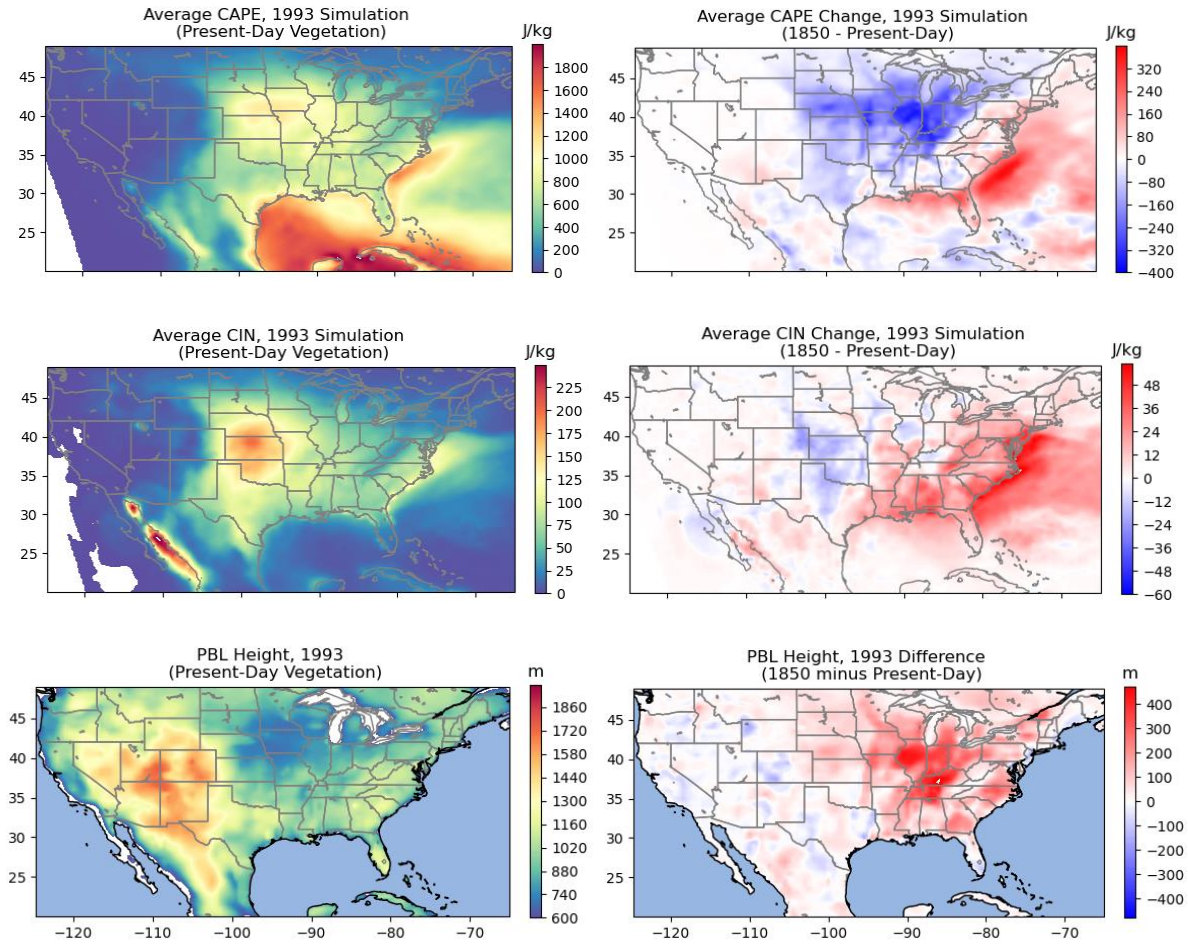


Figure 2.9. Diagnostics of convective triggering for the 1993 simulations. (a) Present-day average convective available potential energy, (b) Convective available potential energy change (1850 minus present-day), (c-d) as for (a-b) for convective inhibition and (e-f) as for (a-b) for PBL height.

To understand the contribution of MCS events to precipitation, we turn to Hovmöller plots showing the time-longitude pattern of precipitation rate, where precipitation was averaged over a latitude band from 37.5 to 45 degrees N. The pattern of increasing longitudinal distance of precipitation features over time suggests the model correctly predicts MCS tracks within the first

25 days, but struggles in capturing the intensity in the second half of our simulations (Figure 2.10). The 1993 present-day vegetation simulation drastically underestimates the amount of precipitation with each MCS. The present-day vegetation simulations underpredict the heavier rainfall events by 70-100% at times. The heavy to extreme rainfall rates were much higher in the observations than the present-day Hovmoller plots. This is a common problem with climate models and low resolution models. The 1850-vegetation simulation was comparable to the present-day vegetation simulation in terms of MCS tracks, but the rainfall intensity was even less. The 1850-vegetation experiment decreased the precipitation during the 1993 events by about 10% relative to the present-day vegetation experiment.

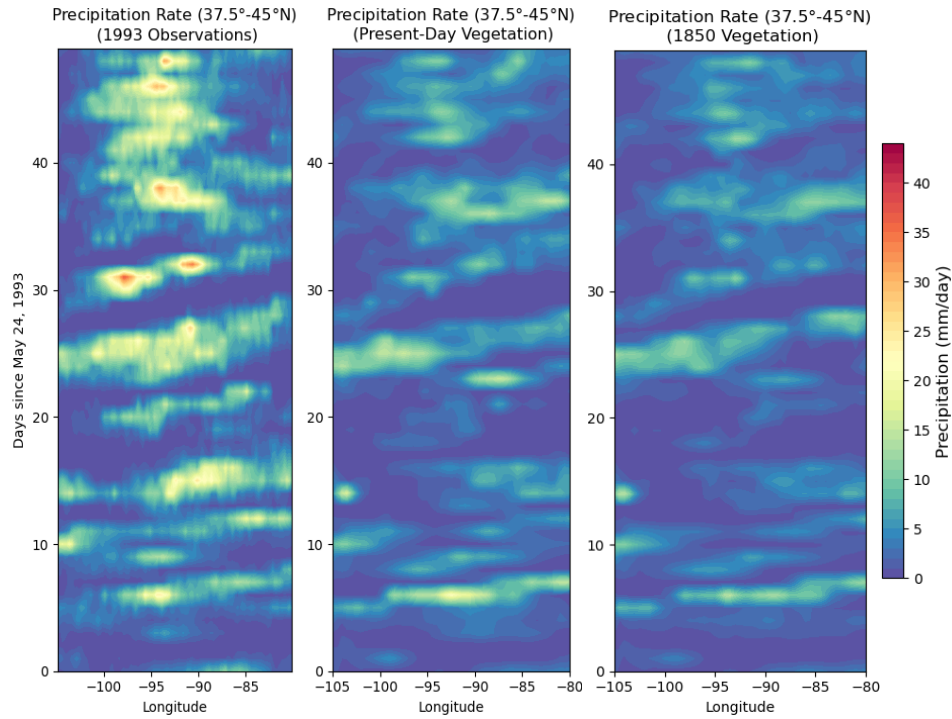


Figure 2.10. Hovmoller plots comparing the precipitation (mm/day) of the present-day vegetation and 1850 vegetation experiments to the observed rainfall from the 50-km 1993 flood case.

If water vapor were the leading factor in determining modeled precipitation responses to 1850 vegetation then we would expect a decrease in precipitation in the 1850-vegetation simulation (Figure 2.11). In fact, precipitation totals in the upper Midwest in 1993 decreased by 15-40% in the 1850-vegetation simulation. Precipitation totals also generally decreased across the Ohio Valley and into Michigan.

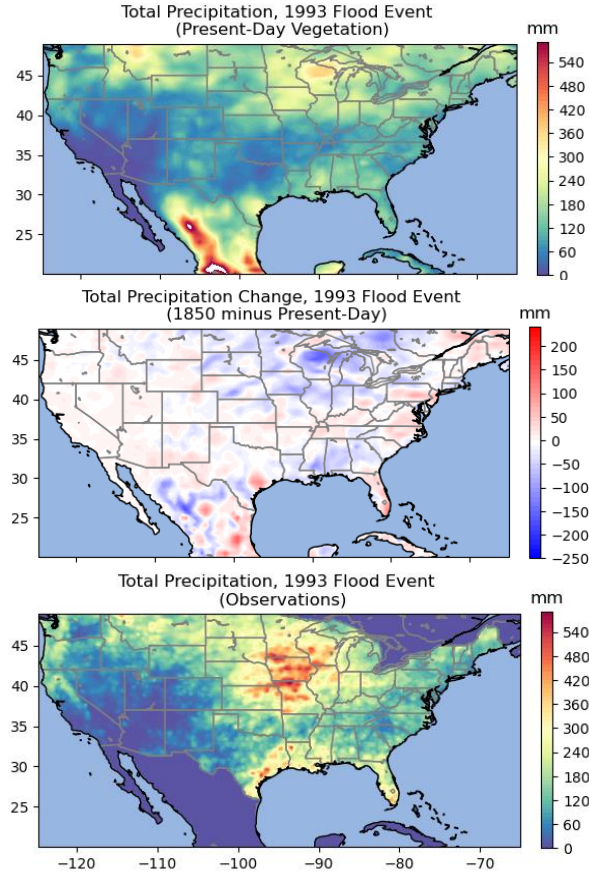


Figure 2.11. The total precipitation (mm) for the 1993 flood case. (a) Present-Day total precipitation (b) Total precipitation change (1850 minus present-day) (c) 1993 observed precipitation.

The modeled precipitation was compared to observations within part of the Midwest (shown in Figure 2.12 with map insert). We calculated the area average for precipitation at each time step and accumulated the precipitation with time. At the end of the 1993 flood event, the present-day vegetation simulation accumulated a regional average precipitation of 134 mm, and the 1850-vegetation simulation accumulated 114 mm, or 85% of the present-day vegetation precipitation. The present-day vegetation experiment for 1993 underestimated rainfall when compared to the observations. The observed total area averaged precipitation was 245 mm. The

present-day rainfall was just 57% of the observed total precipitation when comparing the model output to the CPC observations within our defined area.

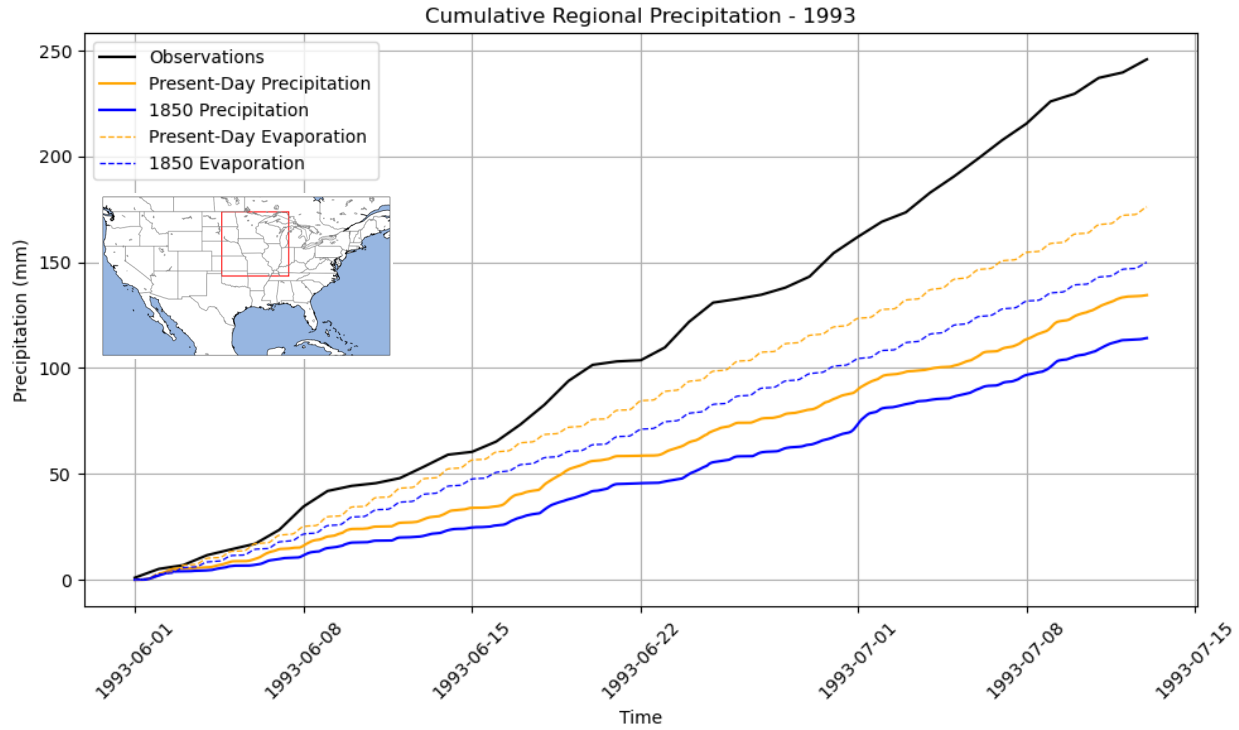


Figure 2.12. 1993 cumulative precipitation (mm) from 36°N to 49°N and 86°W to 100°W. The present-day (orange) diverges from the 1850 precipitation (blue) throughout the simulation period. Accumulated observed precipitation (black) is greater than in either model simulation. Also shown are the surface evaporation for the present-day vegetation (dashed orange) and 1850-vegetation (dashed blue). Precipitation is averaged over the area and then accumulated with each time step.

To further explore the water budget in the Midwest, we calculated precipitation (P) minus evaporation (E) across the defined region from 36°N to 49°N and 86°W to 100°W (Figure 2.13).

Although the present-day vegetation produced greater precipitation it also led to an increase in evaporation rate. For much of the period, the 1850-vegetation experiment P-E is relatively similar to the P-E from the present-day simulation. Around July 1, the P-E in 1850-vegetation jumps higher and separates itself from the present-day vegetation. The 1850-vegetation P-E is -35 mm at the end of the model run. This is 85% of the present-day vegetation P-E total of -41 mm. The largest divergence appears to occur in the first two weeks of June. This is where we see the 1850 surface evaporation is larger than in the present-day in Figure 2.13. At the end of the model run, the 1850-vegetation experiment actually has a larger P-E, meaning that the flooding could have been worse with 1850 vegetation despite lower precipitation. These two experiments have similar P-E because the precipitation is higher in the present-day, but the latent heat flux (evaporation) is also higher.

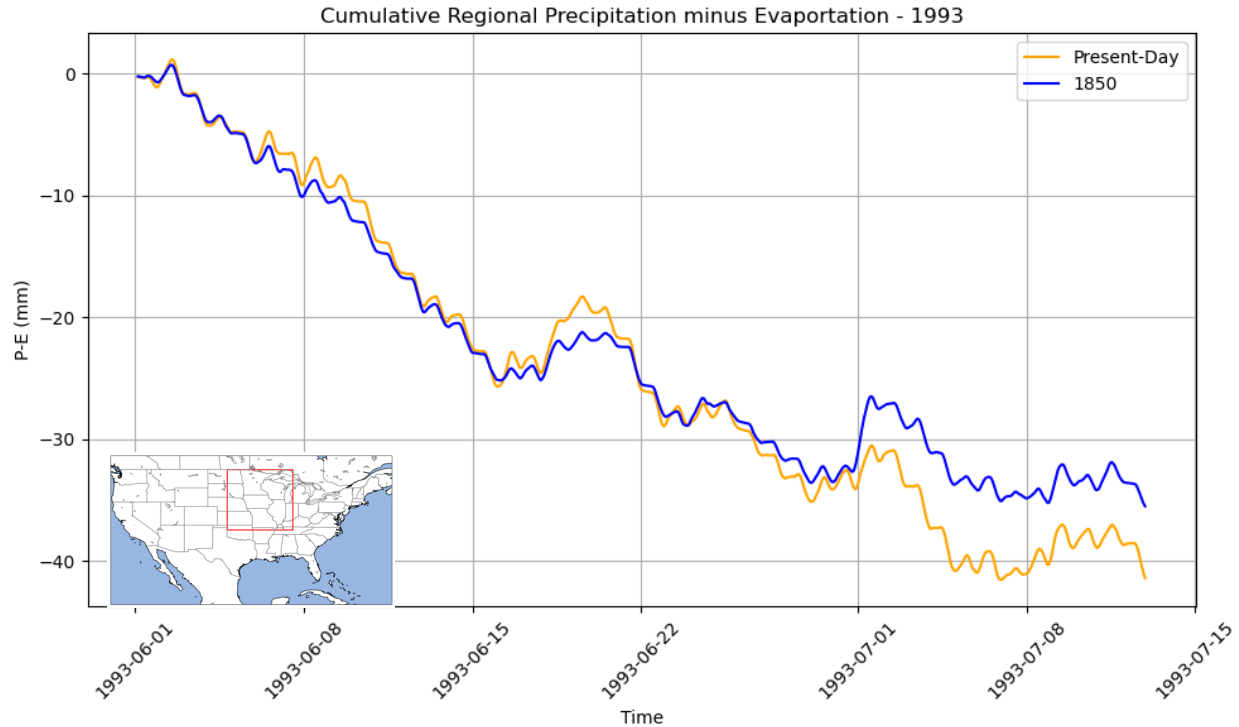


Figure 2.13. 1993 cumulative precipitation (mm) minus evaporation from 36°N to 49°N and 86°W to 100°W. The present-day (orange) diverges from the 1850 precipitation (blue) throughout the simulation period. Precipitation and evaporation is averaged over the area and then accumulated with each time step.

The changes in precipitation extremes were quantified by calculating the empirical probability density function of hourly rain rates. Moderate to heavy rain rates increased in frequency in the present-day experiment while the frequency of 0 rain rates significantly decreased (note the log scale in Figure 2.14). Interestingly, the increase in frequency of heavy rain did not carry over into the extreme tail of the distribution, where there were some instances where extreme rainfall rates occurred with 1850 vegetation but not present-day vegetation, especially over 16 mm. Overall, the distribution of precipitation rates tended to have more precipitation at the lower (0-2

mm/hr) and more comparable to the present-day in the higher and extreme (13+ mm) ranges.

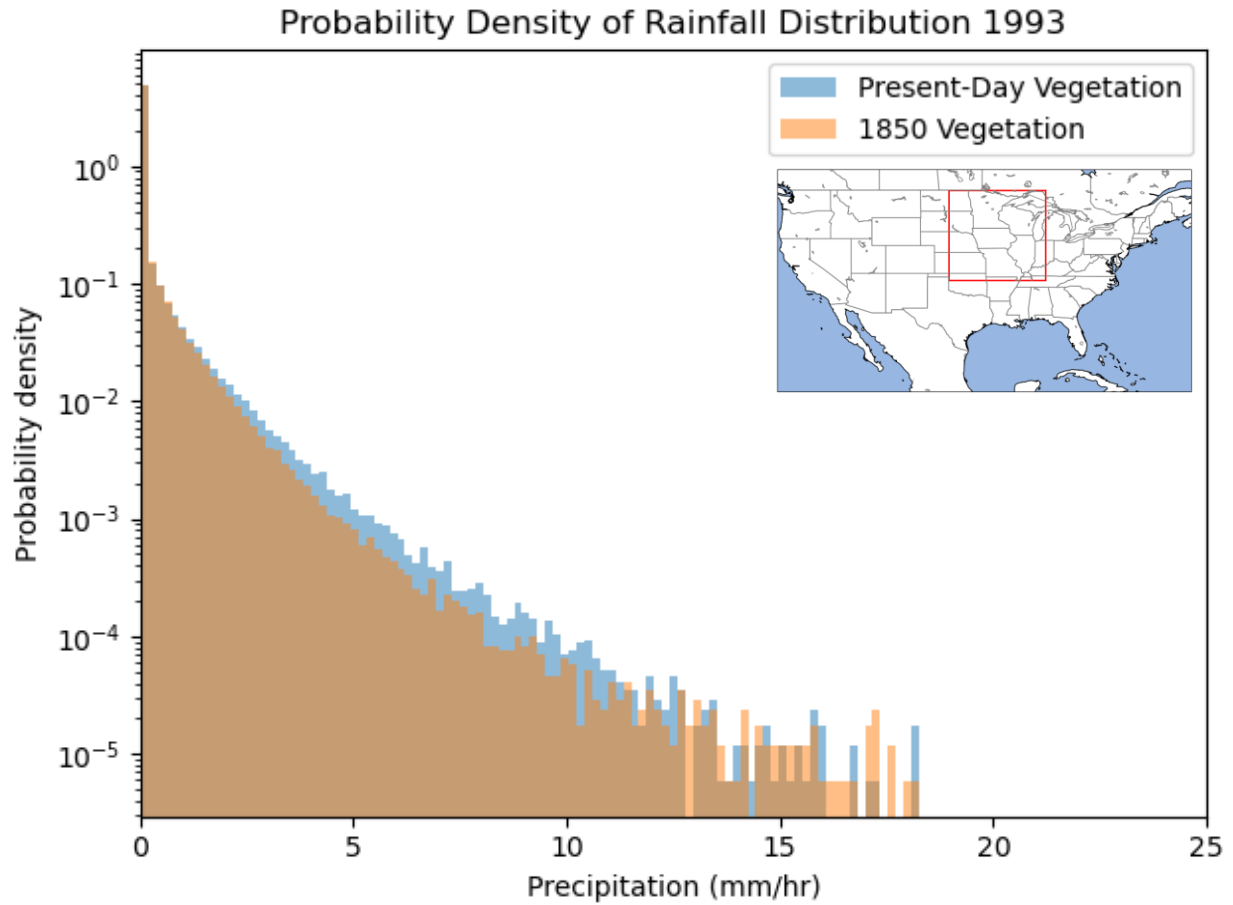


Figure 2.14. 1993 probability density of rainfall for all grid cells from 36°N to 49°N and 86°W to 100°W. Rainfall in the present-day vegetation experiment (blue) diverges from the 1850 vegetation experiment (orange) where there is more frequent moderate and heavy rain in the 3-12 mm range, but there are comparable instances of higher precipitation totals in the 1850-vegetation and present-day experiments above 15 mm.

2.3.2 2008 Flood Event

The 2008 simulations ran from April 24, 2008 to June 13, 2008. A significant difference between this event and the 1993 event is the higher vegetation ELAI early in the simulation for the 1850 vegetation experiment (Figure 2.15), due to the earlier seasonal timing of the floods when crops (dominant in the present-day vegetation experiment) had not yet reached full leaf area. The ELAI was greater in 1850 mostly due to the forests and lack of crop cover compared to the present-day. In general however, the change in ELAI was similar to that of the 1993 simulations shown earlier., with overall greater ELAI in the 1850 vegetation experiment due to forests.

The 1993 and 2008 cases showed similar surface heat flux responses to LULC change. Latent heat flux decreased and sensible heat flux increased in the areas where forests replaced crops in the 1850 vegetation experiment, from the eastern Plains through the upper Mississippi River Valley. Sensible heat flux increased from 0-50 W/m² to 70-130 W/m² with 1850 vegetation (Figure 2.15e, f). Latent heat flux decreased with 1850 vegetation, more so in 2008 than 1993 (Figure 2.15c, d). The increase in sensible heat is consistent with the increase in forest ELAI with 1850 vegetation, similarly to the 1993 simulations.. It may be noted that the sensible heat flux averaged to nearly 0 W/m² in some crop areas, which may reflect an overestimate of the transpiration from crops or excessive nighttime dew formation in the model.

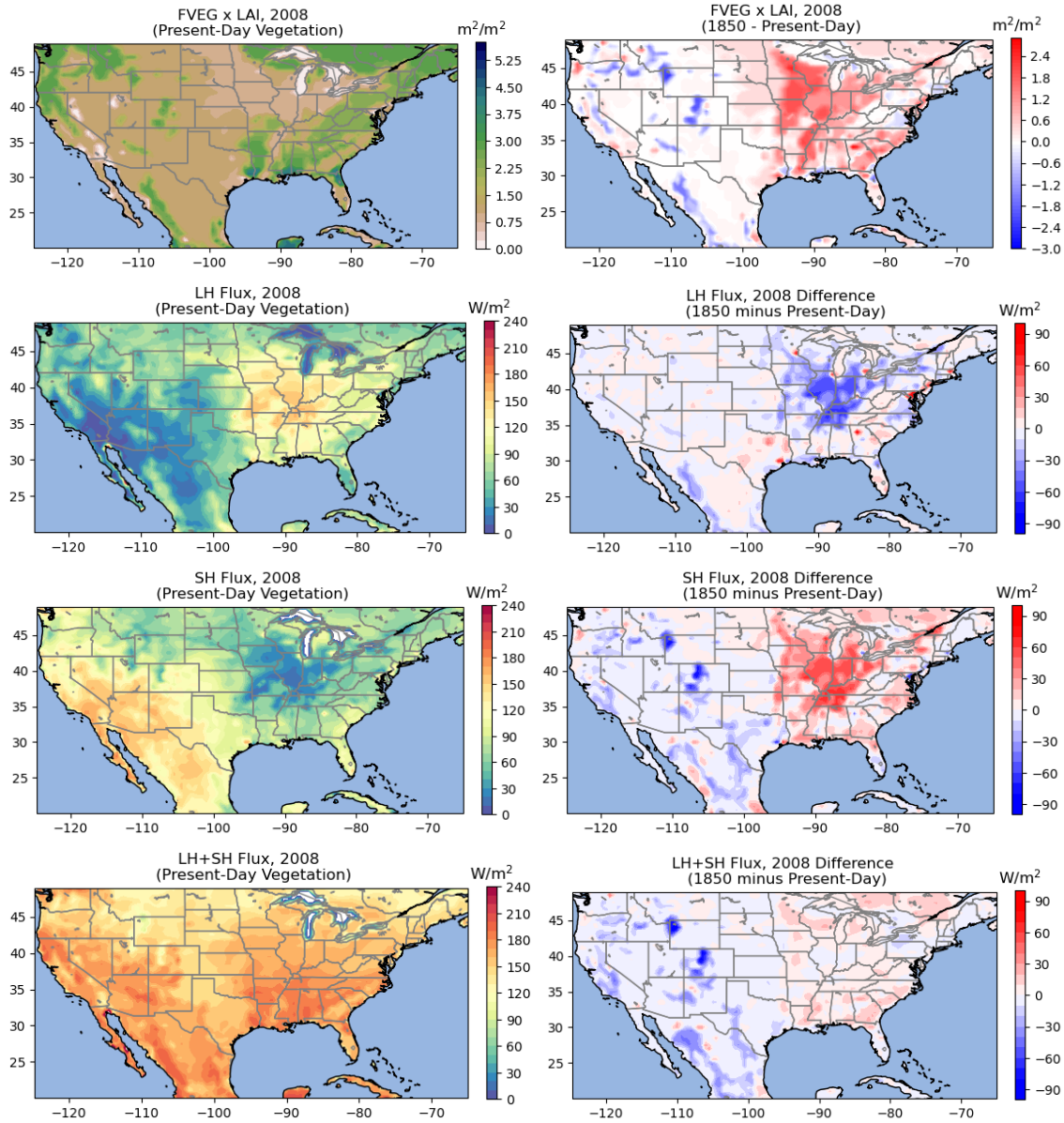


Figure 2.15. Present-day and 1850 vegetation experiments and corresponding change in surface heat fluxes for the 2008 flood case. (a-b) Present-day effective LAI (FVEGxLAI) and change from 1850 (1850 - present-day), (c-d) as for (a-b) for latent heat flux, (e-f) as for (a-b) for sensible heat flux, and (g-h) as for (a-b) for total heat flux (SH + LH).

Albedo decreased by about 30-40% with 1850 vegetation (Figure 2.16a, b). Again, this decrease in albedo could explain the sensible heat flux increases with 1850 vegetation because more

shortwave radiation is being absorbed at the surface due to the 1850 forests and savannas being darker than the present-day croplands. Likewise, the 2-meter air temperature increased in the 1850 vegetation experiment, and was over 15% larger than the present-day temperature in some areas (Figure 2.16c, d). In general, most areas had an increase in temperature anywhere from 1-3°C with 1850 vegetation. Like the 1993 simulations, the results suggest that albedo plays an important role in the change in surface temperature from 1850 to present-day. However, there are some areas of warming (such as the Dakotas) that did not reflect the change in albedo. These areas still saw an increase in ELAI with 1850 vegetation, leading to a decrease in latent heat flux and an increase in sensible heat flux.

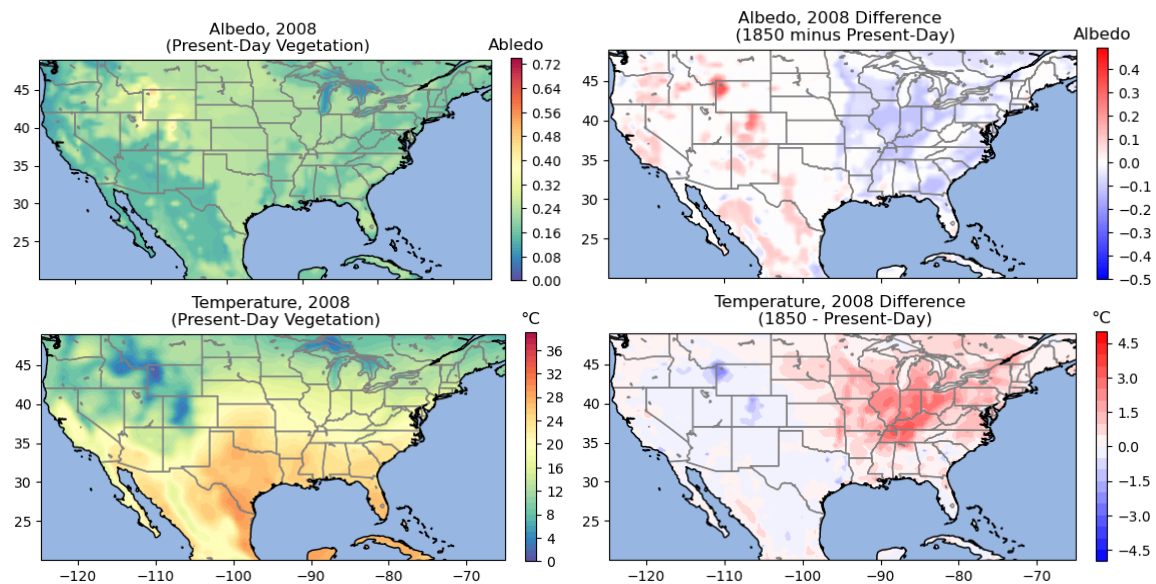


Figure 2.16. Average surface albedo change and 2m air temperature from the 2008 flood case simulations. (a) Present-day average albedo (b) Albedo change (1850 minus present-day) (c) Present-day temperature (d) Temperature change (1850 minus present-day).

The average sea-level pressure was lower in 1850 than the present-day (Figure 2.17a, b). The average sea-level pressure fell by up to 2 mb in over parts of the Eastern US and Atlantic in

response to the 1850 vegetation. Interestingly, the average sea-level pressure increased 1-2mb from the Rockies, through Texas, and westward. The decrease in surface pressure over the eastern US is consistent with a hydrostatic response to the increase in surface temperature, similar to how a thermal low is caused by diverging air at higher levels above a warmer layer. 10-meter wind speeds decreased with 1850 vegetation similar to the 1993 simulations (Figure 2.17c, d). Some areas with the greatest deforestation had a decrease in wind speeds of 50% or greater with 1850 vegetation. As in the 1993 simulations, the decrease in wind speeds was greatest in areas where forests replaced crops in the 1850 vegetation experiment.

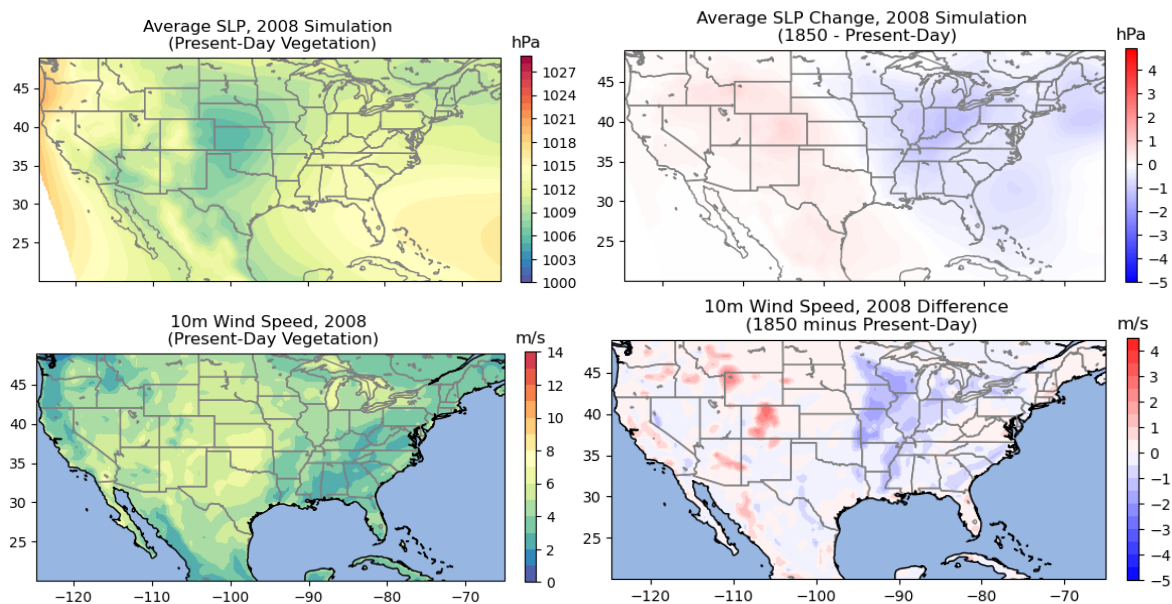


Figure 2.17. Average sea level pressure and 10m wind speed from the 2008 flood case simulations. (a) Present-day average sea level pressure (b) Sea level pressure change (1850 minus present-day) (c) Present-day average 10m wind speed (d) 10m wind speed change (1850 minus present-day).

Water vapor decreased in the 1850-vegetation simulations especially where there was a decrease in latent heat flux, similar to the 1993 results. 2m water vapor mixing ratio decreased across much of the central United States (Figure 18a, b). The precipitable water slightly decreased along the Mississippi River Valley from Louisiana up through the upper Midwest (Figure 18c, d). It is interesting that this decrease occurs where moisture advection is likely to occur from the Gulf of Mexico. Like the 1993 results, this decrease in precipitable water is likely due to decreased moisture advection from a weakened LLJ and decreased latent heat flux at the surface.

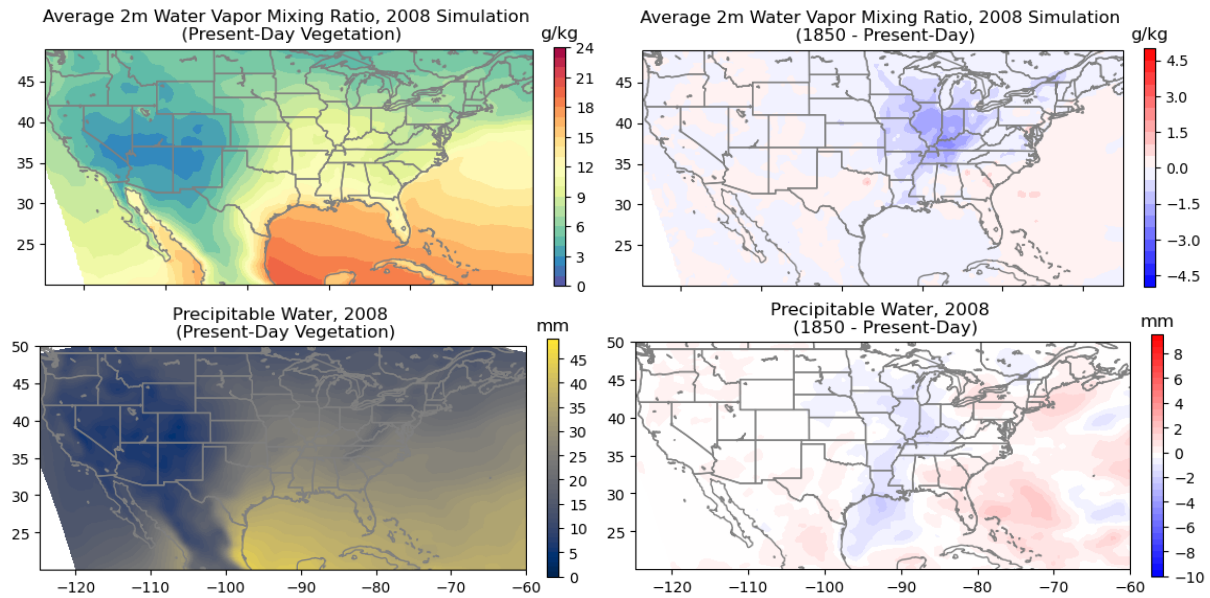


Figure 2.18. Average water vapor variables from the 2008 flood case simulations. (a) Present-day average 2m water vapor (b) 2m water vapor change (1850 minus present-day) (c) Present-day precipitable water (d) Precipitable water change (1850 minus present-day).

The 850mb wind speeds decreased on average in the 1850-vegetation simulations (Figure 2.19a, b). This is noticeable especially where you would expect moisture transport from southerly 850mb winds coming out of the Gulf of Mexico. 850mb wind speeds in 1850 increased off the east coast of Florida, but decreased further out in the Atlantic.

The 850 mb geopotential height decreased in the area of the Bermuda High extension over Florida, and increased in surrounding areas, creating a relaxation of the height gradient in response to 1850 vegetation. This is similar to the changes in the 1993 experiment, and led to a reduction in 850 mb wind speed of about 10 m/s. The 900 mb wind speed showed similar changes, again pointing to a reduction in intensity of the Great Plains low level jet.

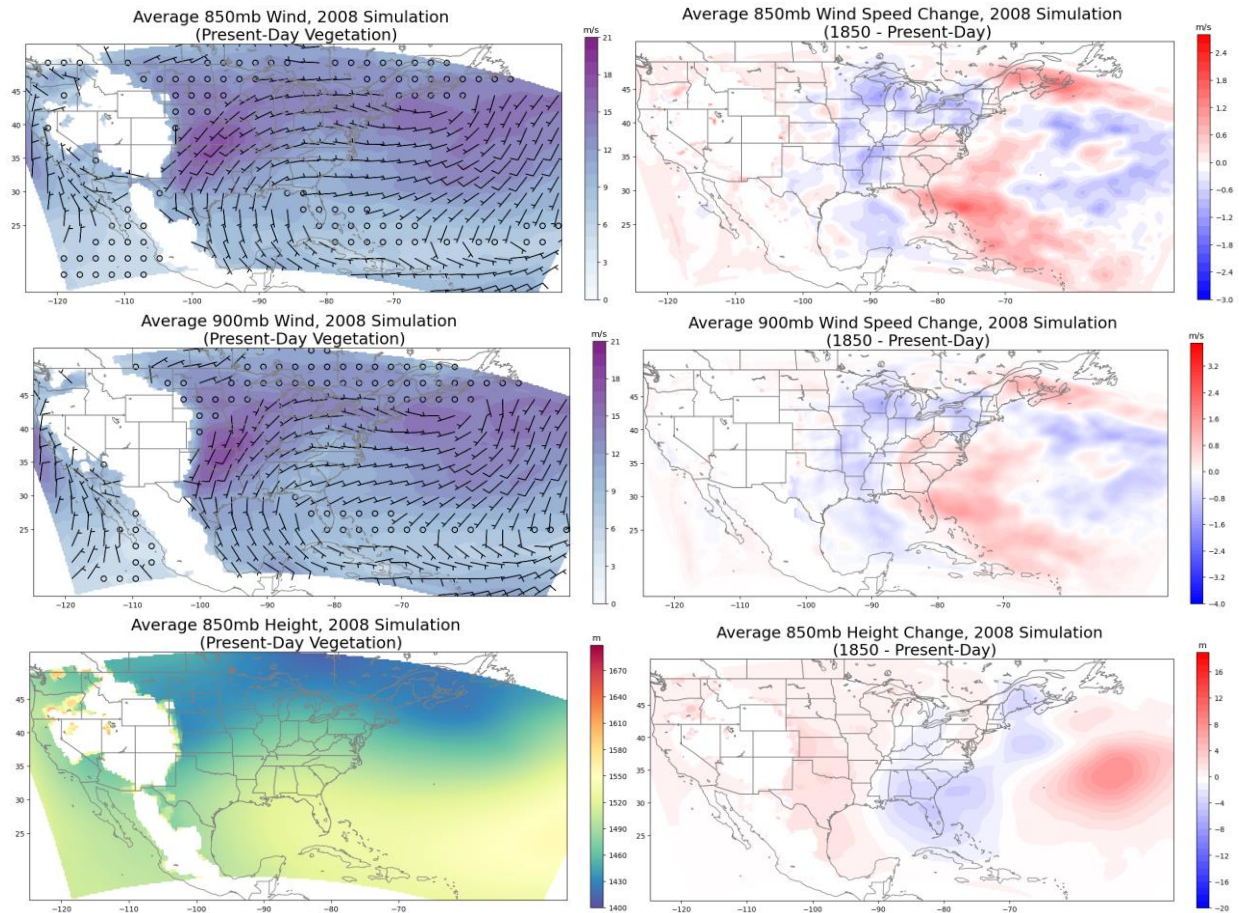


Figure 2.19. Responses of low-level wind speed and geopotential heights in the 2008 experiments. (a) Average 850 mb wind speed (m/s) and direction (m/s) with present-day vegetation, (b) 850 mb wind speed change (1850 minus present-day vegetation). (c-d) as in (a-b) but for 900 mb wind speed (m/s) and direction (m/s) (e-f) As in (a-b) but for 850 mb height.

In the 2008 case, CAPE decreased in response to 1850-vegetation by nearly 200 J/kg over a large part of the Midwest (Figure 2.20a, b), with a decrease of nearly 25% in Illinois. This is a similar change to the 1993 simulations. The decrease in CAPE may have contributed to the decrease in precipitation, especially with a CAPE-sensitive convection scheme. CIN also decreased over much of the central US in the 1850-vegetation experiment. A decrease in CIN would allow more convection to occur over larger areas, but with less intense rainfall. Like 1993,

the changes in convective inhibition (CIN) were more spatially variable than those of other parameters from 1850 to present-day (Figure 2.20c, d). CIN decreased in the 1850-vegetation simulations over the southern and central Plains. The CIN also decreased along the Mississippi River Valley up to the northern Illinois border. The pattern of decrease in CIN is similar to that of wind speed, which suggests an influence of the LLJ and large-scale advection. PBL heights in the present-day were 600-900 meters and increased by 200-400 meters with 1850 vegetation. This is a 20-50% increase in PBL height. The PBL heights in the 2008 present-day are lower than the PBL heights in 1993, but there is a larger percent increase in PBL heights in 2008 (Figure 2.20e, f). Like the 1993 case, this increase in PBL height is due to the surface temperature increase from increased sensible heat flux at the surface.

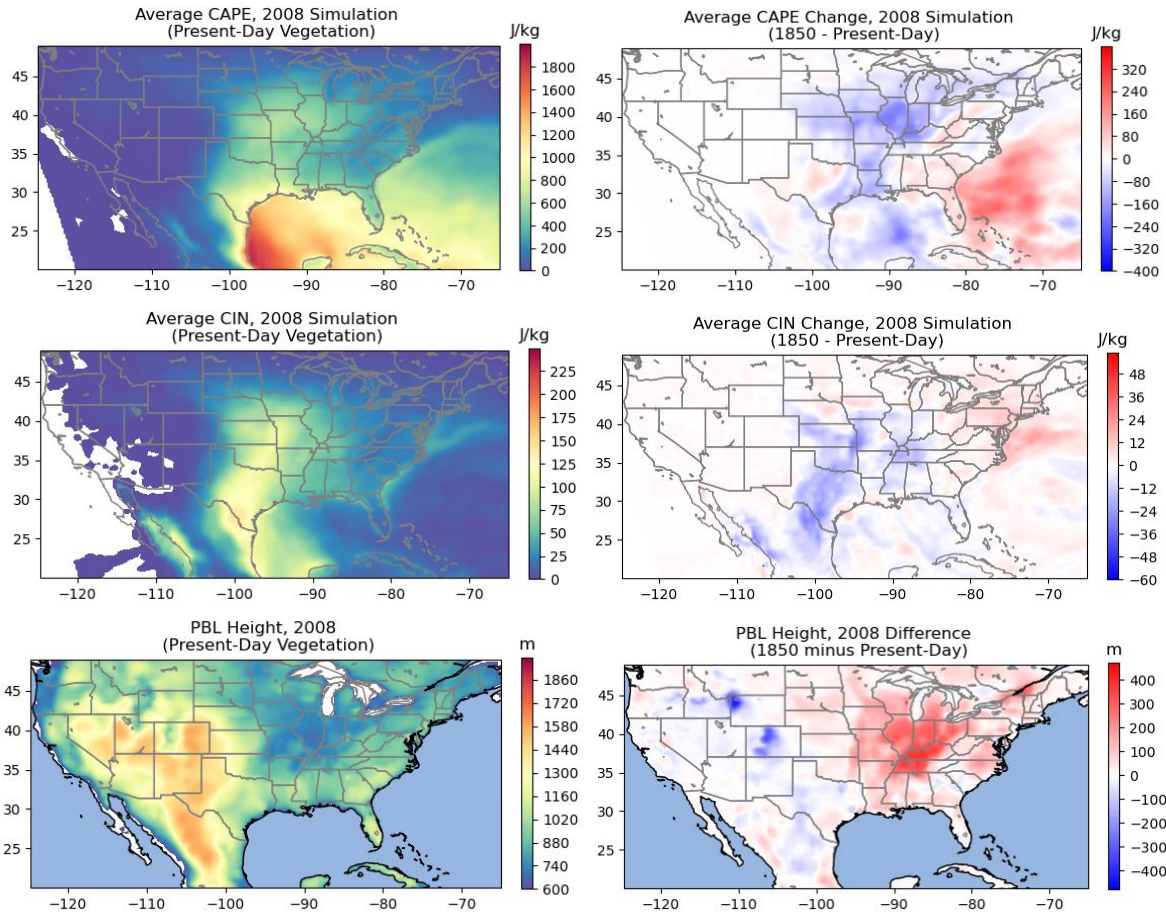


Figure 2.20. Average thermodynamic variables for the 2008 simulations. (a) Present-day average convective available potential energy (b) Convective available potential energy change (1850 minus present-day) (c) Present-day average convective inhibition (d) Convective inhibition change (1850 minus present-day) (e) Present-day average PBL height (f) PBL height change (1850 minus present-day).

Like the 1993 simulation, the 2008 present-day vegetation experiment underestimated rainfall compared to the observations. The simulated rainfall was 84% of the observed total precipitation within the defined region (shown in Figure 2.21) where the total area averaged observed rainfall was 213 mm. The underprediction of total rainfall from the observations to the

present-day vegetation simulation could be a product of the convective scheme or microphysics scheme in the model. The present-day vegetation experiment accumulated an area average precipitation of 135 mm, compared to 117 mm in the 1850 vegetation experiment, giving a 13% decrease with 1850 vegetation. It should be noted that there was an increase in cumulative evaporation in the present-day vegetation experiment (orange dashed line).

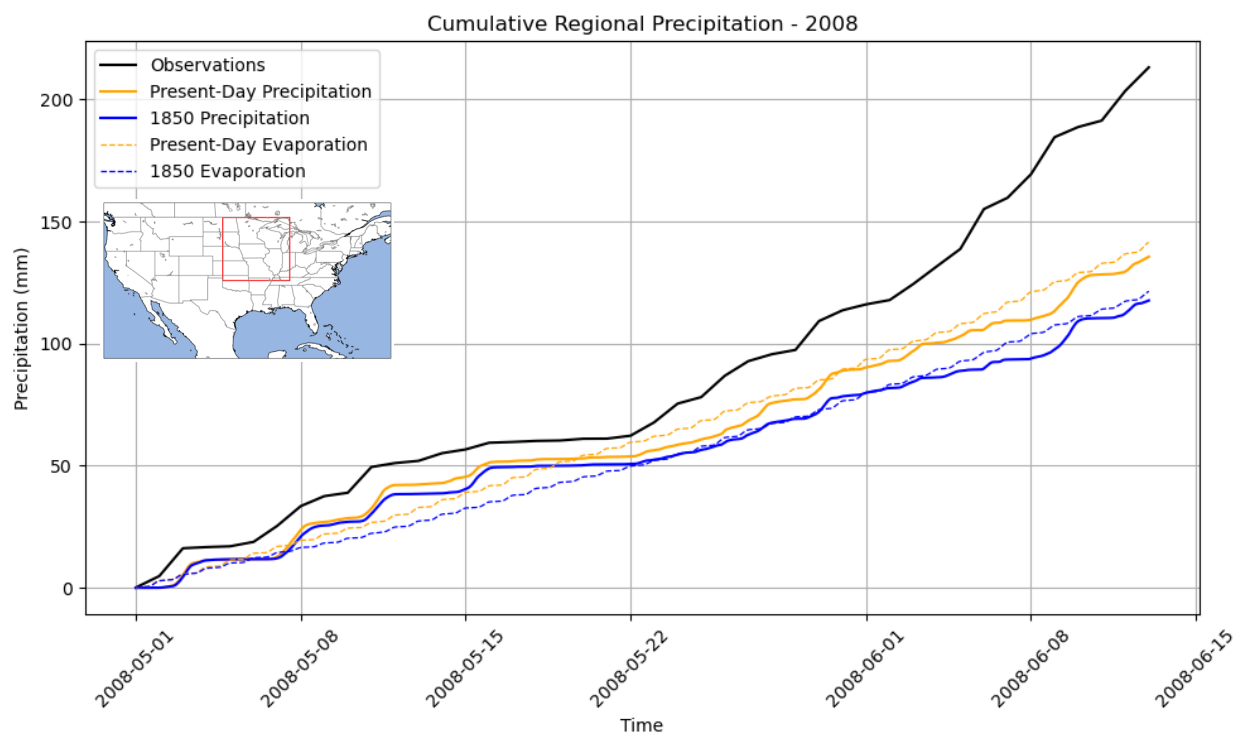


Figure 2.21. 2008 cumulative precipitation (mm) from 36°N to 49°N and 86°W to 100°W. The present-day (orange) diverges from the 1850 precipitation (blue) throughout the simulation period. Observed precipitation (black) accumulates much more than either model simulation. The surface evaporation for the present-day vegetation (dashed orange) and 1850-vegetation (dashed blue) Precipitation is averaged over the area and then accumulated with each time step.

Again, we further explore the water budget in the Midwest by looking at the precipitation minus evaporation across a specified domain. Similar to the 1993 simulations, P-E shows that surface water is much less variable than precipitation, meaning evaporation is overall keeping up with the precipitation in our simulations. As in the 1993 flood event, this suggests that the lower evaporation rate in the 1850-vegetation experiment is playing a part in the flooding. In the middle period, around 24 May, the 1850 vegetation experiment showed an excess of 5 cm or nearly 2 inches of water compared to the present-day vegetation experiment, due to the lower evaporation rate.

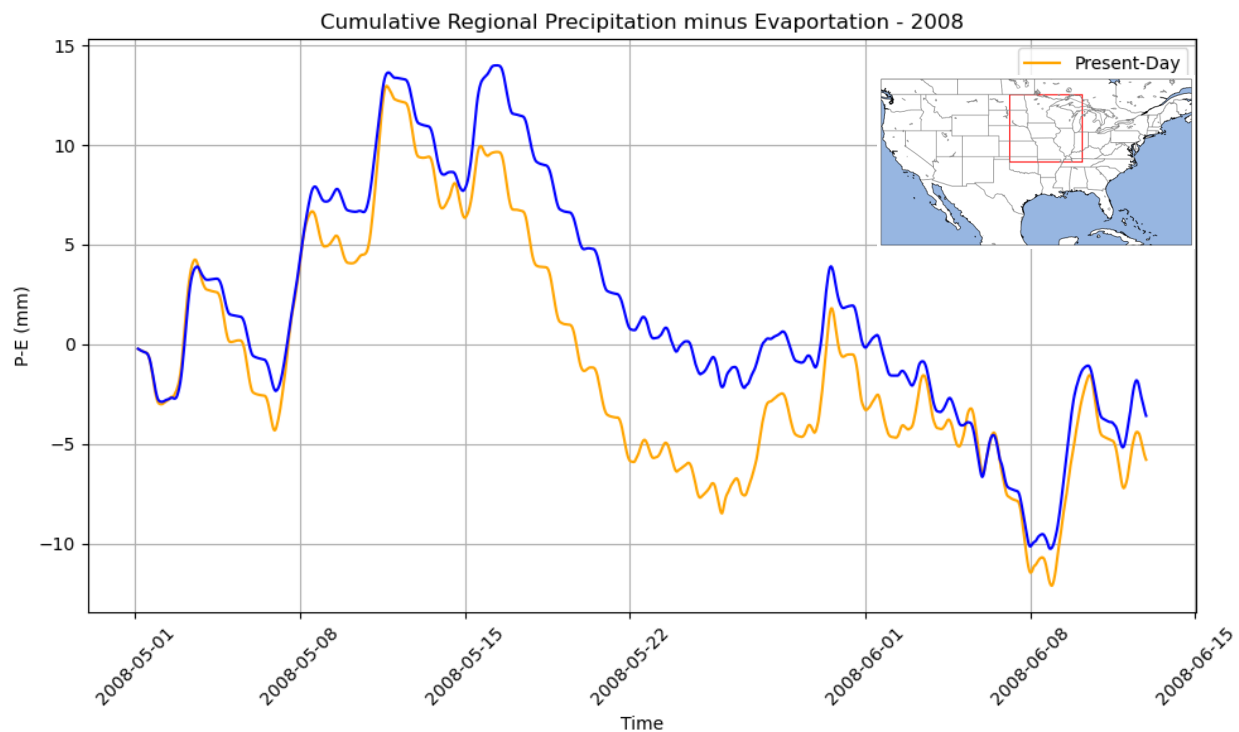


Figure 2.22. 2008 cumulative precipitation (mm) minus evaporation from 36°N to 49°N and 86°W to 100°W, for the present-day vegetation experiment (orange) and the 1850 experiment (blue). Lower evaporation rates lead to more surface water (P-E) in the 1850 experiment despite lower rainfall rates.

The probability density plot of rainfall in 2008 shows the increased probability of moderate to heavy rainfall in the present-day vegetation experiment. The skewed shape of the differences indicates that heavy precipitation in the present-day vegetation experiment is more likely than in the 1850 experiment, with an increase in extreme rainfall due to the present-day vegetation.

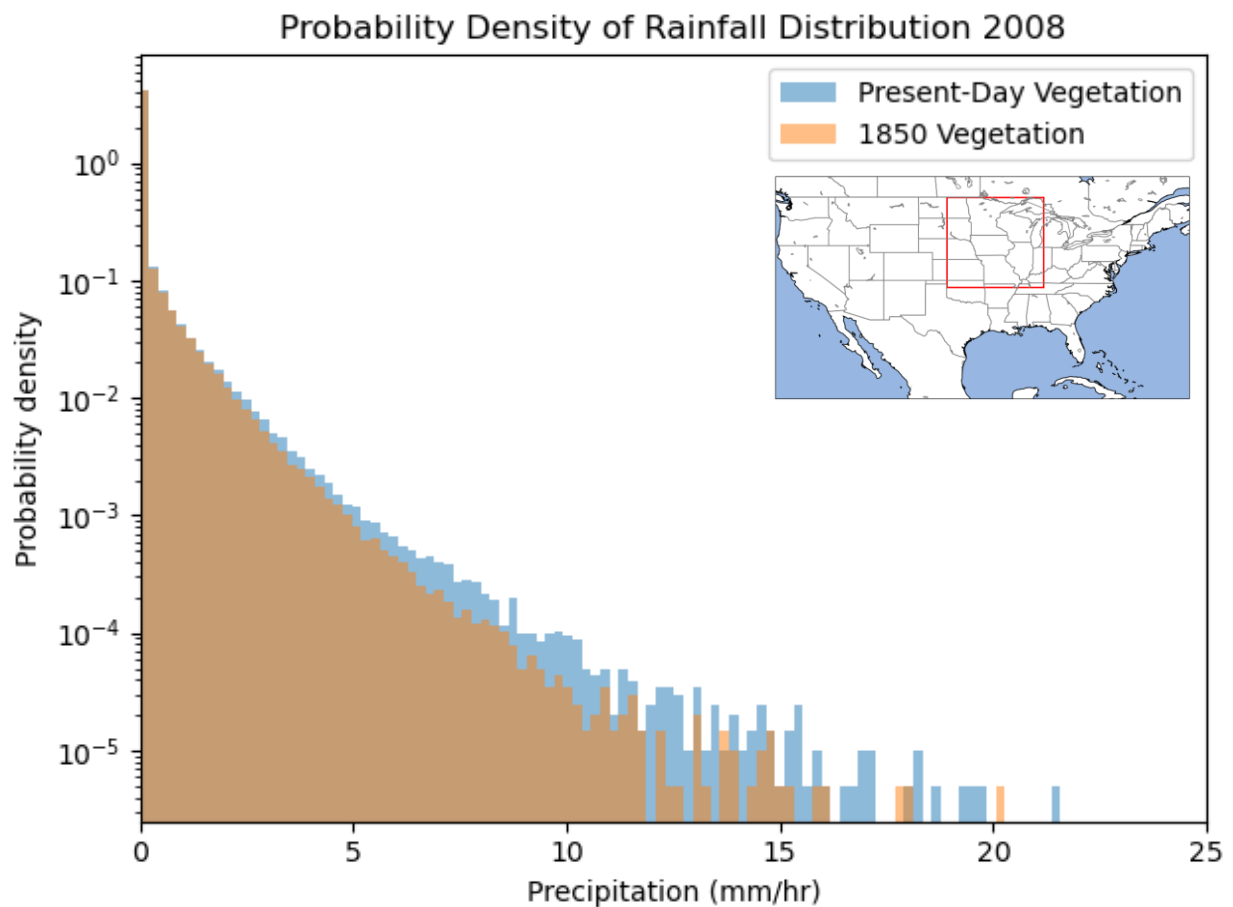


Figure 2.23. 2008 probability density plot of rainfall distribution from 36°N to 49°N and 86°W to 100°W. The present-day vegetation experiment (blue) diverges from the 1850 experiment (orange), especially at higher rain rates.

The Hovmoller plot for 2008 shows that the hindcast predicts the MCS tracks well through about day 25 (Figure 2.24). The 1850-vegetation simulation was comparable to the present-day vegetation simulation for MCS tracks, but the rainfall intensity was less. However, this bias was not as severe as the 1993 simulations, especially in the beginning of the event. The 2008 present-day vegetation simulation also underestimated the amount of rainfall with each MCS, particularly toward the middle and end of the event.

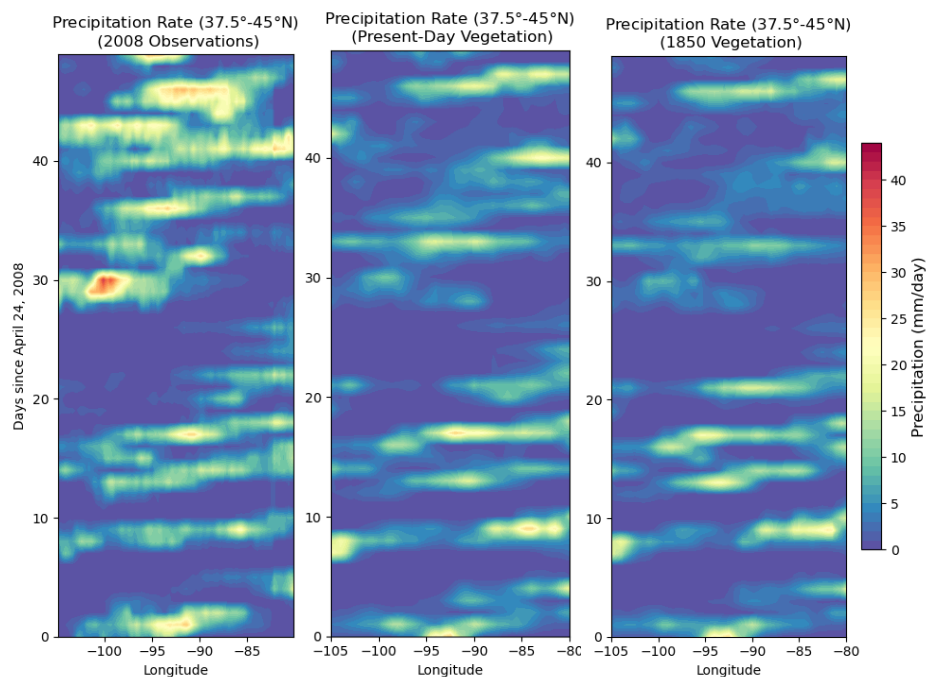


Figure 2.24. Hovmoller plots comparing the precipitation (mm/day) of the present-day vegetation and 1850 vegetation to the observed rainfall from the 50-km 2008 flood case.

Similar to the 1993 simulation, the precipitation in the 1850-vegetation simulations decreased across much of the regions where there was a change in ELAI. There was a decrease

of 50 mm or larger in precipitation over eastern Iowa, Illinois, Kentucky, and Wisconsin (Figure 2.25). This further suggests that the change in latent heat flux and precipitable water in these regions had an effect on the precipitation. The simulations struggled to produce the extreme precipitation in Iowa, Wisconsin, and Kansas.

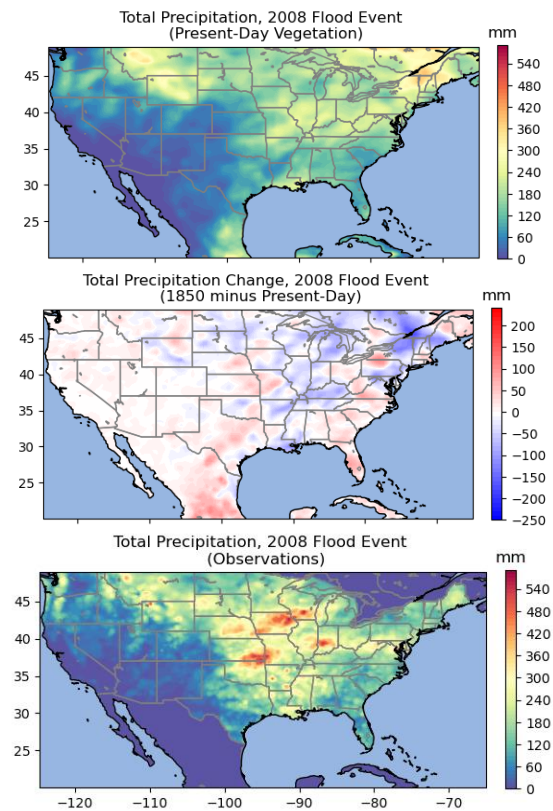


Figure 2.25. The total precipitation (mm) for the 2008 flood case. (a) Present-Day total precipitation (b) Total precipitation change (1850 minus present-day) (c) 1993 observed precipitation.

In summary, for the 1993 and 2008 event averages, the sensible heat flux increased where ELAI increased with 1850 vegetation. The latent heat flux decreased in these same areas. This change may have been led by a decrease in albedo and increase in surface temperature where ELAI increased. The increase in temperatures led to a decrease in the pressure gradients across the domain. The decrease in pressure gradients can weaken the LLJ and slow the moisture advection. Water vapor at the surface decreased with the decrease in latent heat flux. Precipitable water also decreased with 1850 vegetation and may be a result of decreased moisture transport and decreased latent heat flux.

The underestimation of simulated precipitation totals could be due to the coarse resolution and convective scheme. The present-day vegetation experiment underpredicted precipitation by 25-35%. Key results of the present-day and 1850-vegetation experiments were similar between the 1993 and 2008 cases. Both cases showed decreased precipitation totals with 1850 vegetation compared to present-day vegetation. These results suggest that LULLC had effects on the storm environment that ultimately reduced rainfall in the 1850-vegetation experiments.

2.3.3 Case Study: 8 July 1993

In this section we investigate one event identified from the Hovmoller plot. It is difficult to examine MCS structure at 50 km resolution and with a convective scheme. Instead, we will attempt to examine the environments for convection and their differences between the two experiments. It should be kept in mind that direct comparisons between parameterized and observed convection cannot be made due to the lack of realistic representation of convective processes in the parameterizations. For the case study we chose July 8, 1993 because it is a well

documented event due to its severity. This event was a large part of the 1993 flooding in Iowa and caused severe flooding in western Iowa. Quasi-stationary convection occurred on July 8, 1993 which led to extreme rainfall amounts and flooding in the Midwest. Observations show extreme precipitation amounts over Iowa and Missouri with amounts over 160 mm in just 2 days (Figure 2.27d).

The 1993 surface analysis shows that there is a surface low in eastern Colorado (Figure 2.26a). Our model also has a surface low near this general region. Our present-day model shows this surface low in central/western Nebraska (Figure 2.26b). The observed surface analysis shows a stationary front from the low in eastern Colorado through Kansas and Iowa where quasi-stationary convection was observed.

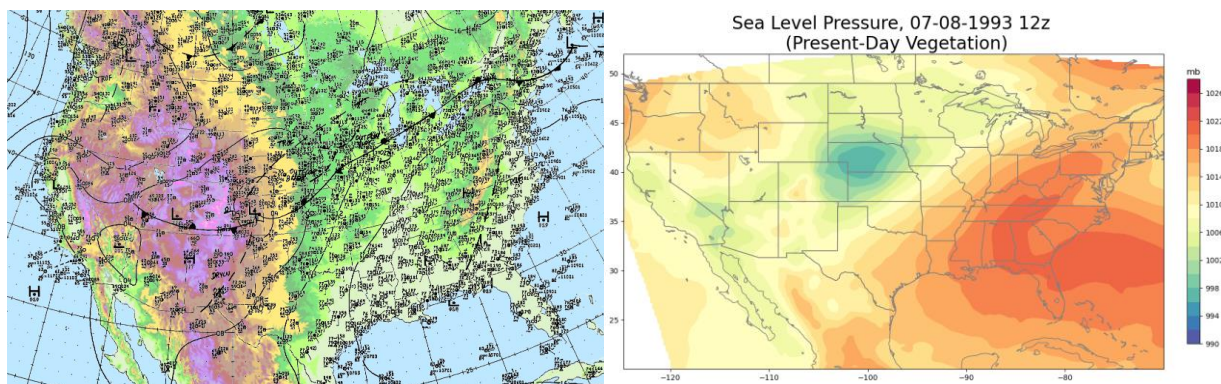


Figure 2.26. 8 July 1993 12z observed and modeled surface analysis. (a) 1993 surface analysis (b) modeled sea level pressure.

The model struggled to produce the extreme precipitation amounts in the Midwest. The observed maximum in Missouri was not reproduced in the model. The precipitation maximum over Iowa in the present-day experiment was 51 mm and did not significantly change in the

1850-vegetation experiment (Figure 2.27a, b). Although maximum precipitation amounts did not change, the mean precipitation amounts across the Iowa, Minnesota, and Missouri region decreased in the 1850-vegetation model from 17 mm to 14 mm. The location of the maximum precipitation shifted slightly west in the 1850 model and appears to not cover as large of an area (Figure 2.27b). The heavy rainfall in the present-day experiment covers a larger area. This change is subtle, but noticeable when comparing area average precipitation from the event.

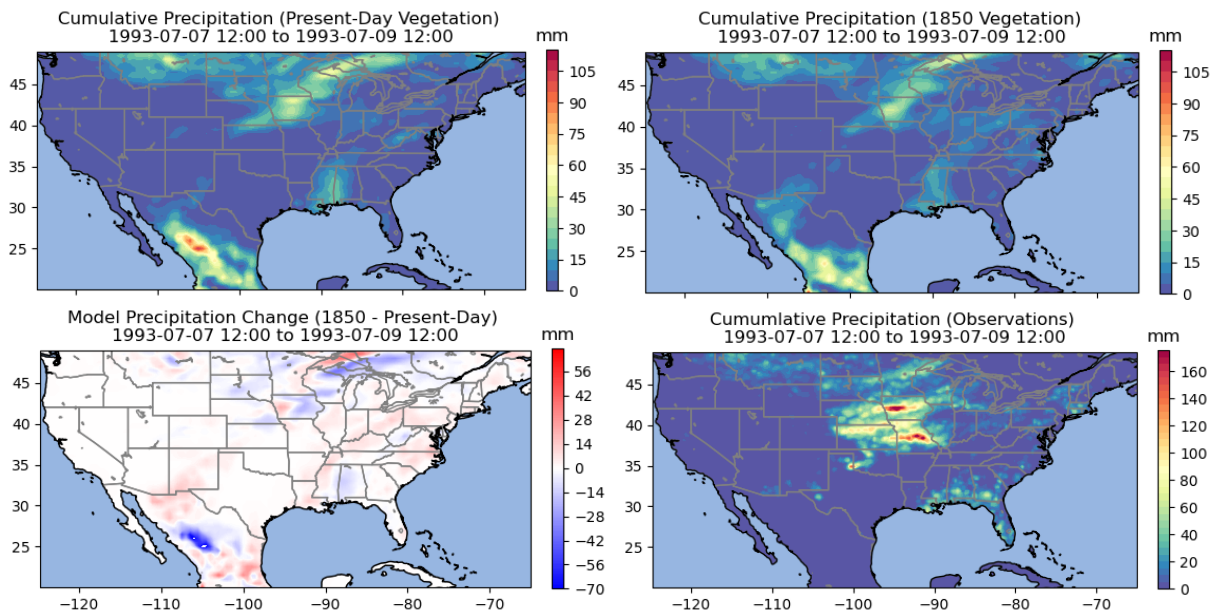


Figure 2.27. 8 July 1993 case accumulated precipitation. Accumulated precipitation (mm) from 12:00 UTC on July 7, 1993 to 12:00 UTC on July 9, 1993 for (a) the present-day vegetation and (b) 1850-vegetation experiment (c) Event total precipitation change (1850 minus present-day vegetation experiments) (d) Observed event rainfall from CPC. CPC observed precipitation begins from 12:00 UTC the day prior, thus we look at day 1 12z to day 3 12z to capture the event.

The 1850 vegetation experiment yielded lower area averaged rainfall rates for nearly the entire event. This could be caused by the weakened LLJ and decreased moisture content. At the peak of the event, the present-day vegetation gave a higher rate of hourly rainfall by about .05 mm/hr.

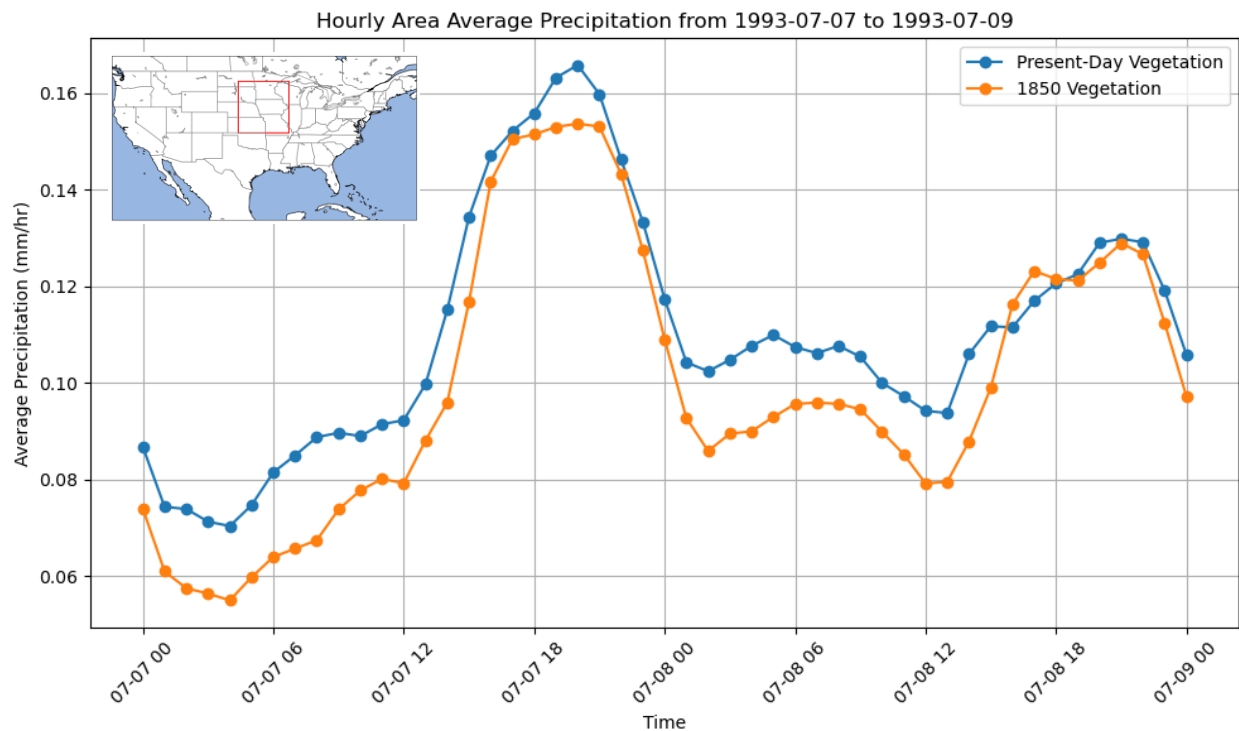


Figure 2.28. 8 July 1993 case average hourly precipitation for the defined region. Rainfall rates from 07-07-1993 00:00 UTC to 07-09-1993 00:00 UTC from present-day vegetation experiment (blue) and 1850-vegetation experiment (orange).

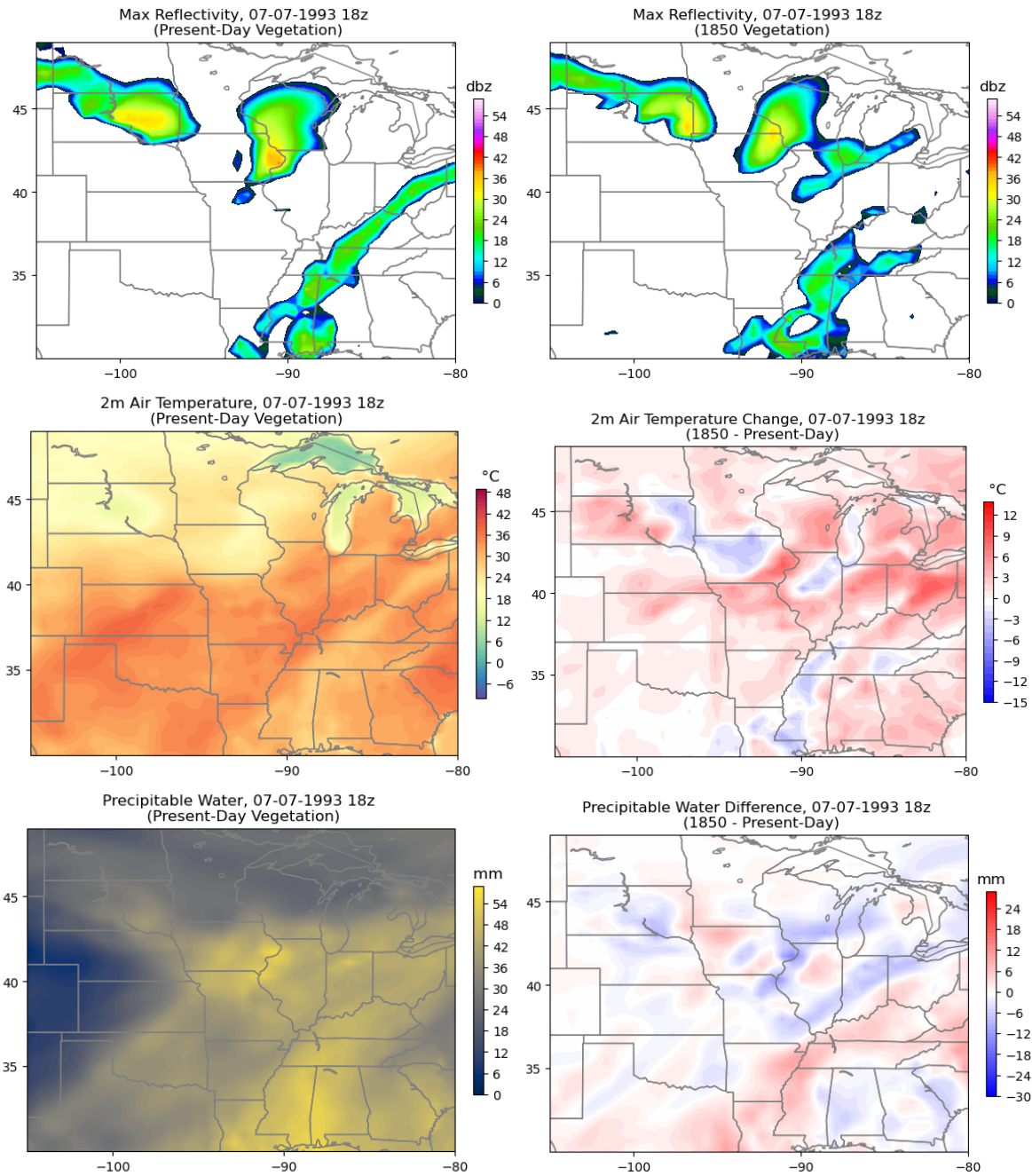


Figure 2.29. 8 July 1993 MCS event. (a) Present-Day reflectivity (dBz), (b) 1850 reflectivity, (c) Present-day temperature (°C), (d) Temperature change (1850 minus present-day), (e) Present-day precipitable water (mm) (f) Change in precipitable water (1850 - present-day).

The temperature in this region increased significantly in the 1850-vegetation experiment. The temperature increased 1-6°C across Nebraska, Missouri, and Illinois. We expect an increase in surface temperature with a decrease in albedo in the 1850-vegetation simulation. This increase in temperature is across the entire domain, so this does not strengthen the front. Precipitable water decreased slightly in the 1850-vegetation simulation, although the changes were more spatially variable in this case compared to some of our other cases (not shown).

The 10 m wind speeds were strong and widespread in the present-day experiment. The surface wind speeds decreased significantly in the 1850-vegetation simulation, especially in Illinois and Indiana where forests were replaced by crops in the present-day. The surface wind speeds also decreased at the surface wind maximum in Kansas where we expect the transport of warm air and moisture. Stronger surface winds can enhance the front by increasing convergence, creating lift, and increasing the temperature gradient.

The 850 mb wind speed maxima were located over Oklahoma and Kansas (Figure 2.30a). This strong (20 m/s) LLJ is transporting warm moist air to fuel the quasi-stationary convection. The low level jet also contributes to convergence and ascent along the front which could enhance updraft strength. The 850 mb wind speeds decrease significantly from Texas, through the 850 mb wind speed maxima in OK/KS, and through Iowa. This decrease in 850 mb wind speeds could be the cause of the slightly decreased precipitation amounts in the region.

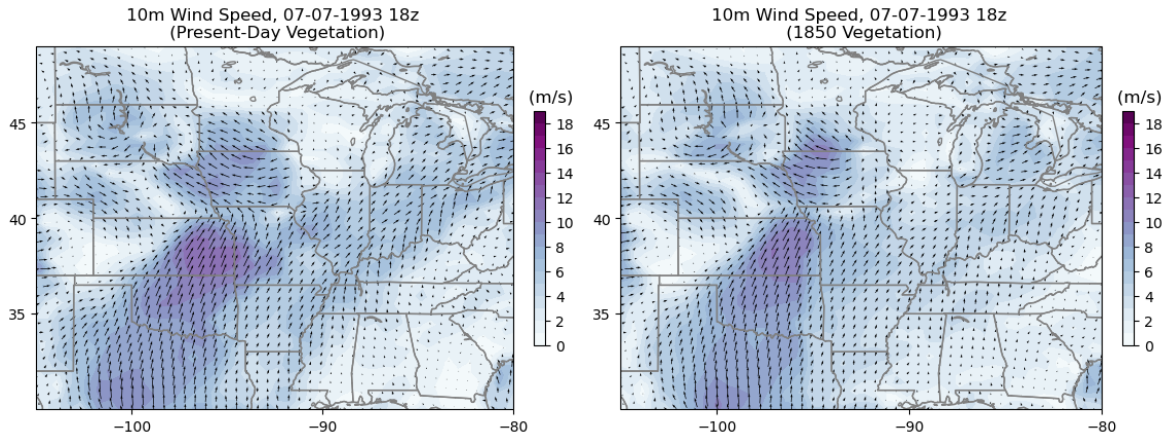


Figure 2.30. Wind speed (m/s) and direction during the 07-07-1993 and 07-08-1993 MCS event at 18:00 UTC. (a) 10 m winds in the present-day vegetation experiment (b) 10 m winds in the 1850-vegetation experiment.

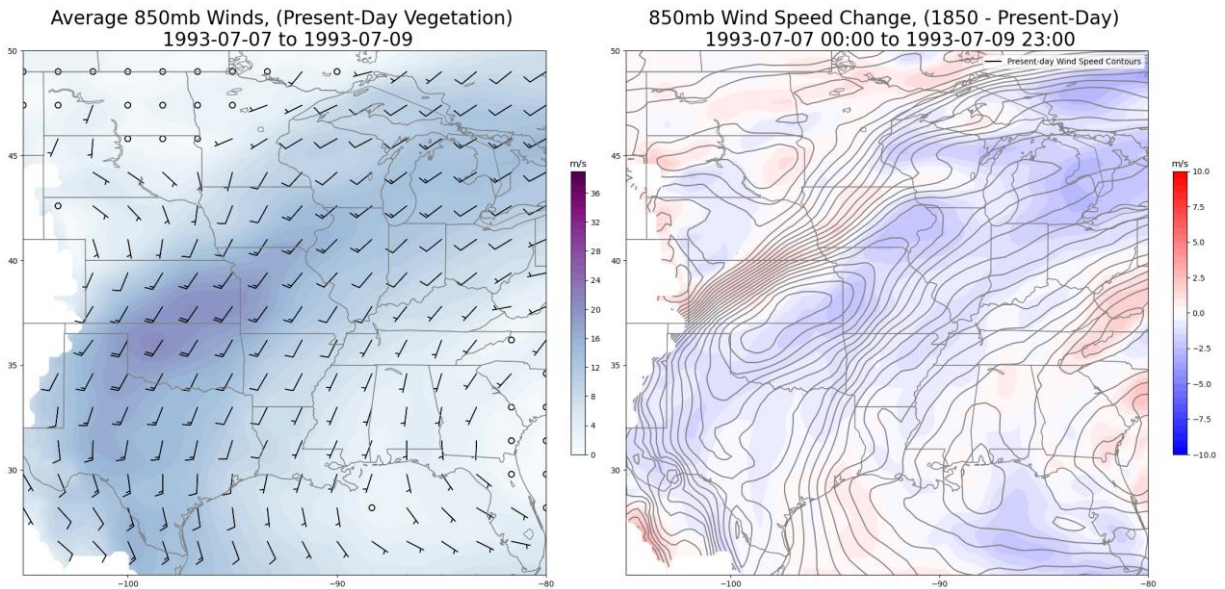


Figure 2.31. Average 850 mb wind speed and direction from July 7, 1993 to July 9, 1993. (a) Average 850 mb winds in the present-day vegetation experiment (d) Change in 850 mb wind speed (1850-vegetation minus present-day vegetation experiments) including the present-day vegetation 850 mb wind speed contours in gray for reference.

The average CAPE is about 16% lower in the 1850-vegetation experiment (Figure 2.32). The maximum CAPE during this time period is about 9% lower in the 1850-vegetation experiment, or about 500 J/kg lower than the 4700 J/kg in the present-day model. Average CIN increased slightly in the 1850-vegetation experiment, possibly suggesting why the precipitation maxima was concentrated in a smaller area. However, the maximum CIN during this time period was higher in the present-day vegetation model. The 2m water vapor mixing ratio decreased in the 1850 vegetation experiment by over 5% (Figure 2.32b).

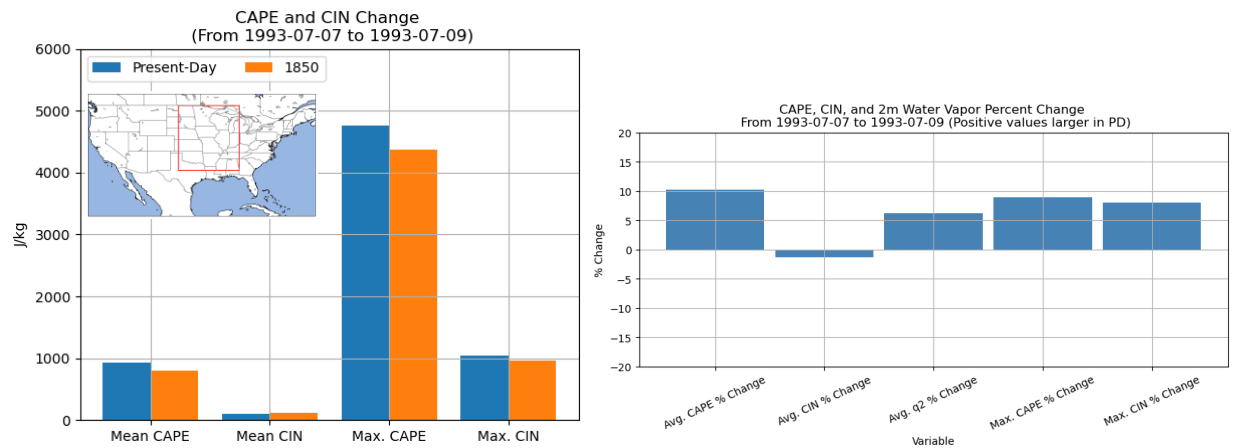


Figure 2.32. 8 July 1993 changes to CAPE and CIN in the defined region 32°N to 49°N and 85°W to 101°W. (a) Present-day and 1850-vegetation area average CAPE, present-day and 1850-vegetation area averaged CIN, present-day and 1850-vegetation area maximum CAPE, present-day and 1850-vegetation area maximum CIN, and (b) Percent change in the same variables as (a) including 2m water vapor mixing ratio.

2.3.4 Additional Cases Studied

Our three other selected cases were June 8, 1993, June 24, 1993, and May 06, 2008. These cases showed similar results to the July 7-9, 1993 event. We will highlight the similarities between the three other cases in relation to the July 8, 1993 case.

The most consistent response among the cases was the 2 meter air temperature increase in the 1850-vegetation experiments in the environment of storms. This warming was especially prominent in regions where ELAI increased and sensible heat flux increased in the 1850-vegetation scenario. Warm air advection may also play a role in increasing the surface temperature as the southerly winds are stronger in these cases. The May 06, 2008 case was an outlier for temperature, however this event occurred in the early morning hours and diurnal heating had not yet occurred.

Precipitable water appeared to decrease in the 1850-vegetation simulations for each of these cases. This was more spatially variable in some cases. This slight decrease in precipitable water could be due to a weakened LLJ and decreased surface latent heat flux in the 1850-vegetation simulations. A slight decrease in precipitable water is what we might expect when comparing to the averages for the flood events, where there was an overall decrease in average precipitable water across much of the domain in the 1850-vegetation experiment. Likewise, the 2 meter water vapor mixing ratio also decreased in all of our cases in the 1850-vegetation simulations.

The 10 meter wind speeds decreased in the 1850-vegetation simulations in all cases. The increased roughness length from the savannas and forests in 1850 will cause the surface wind speeds to slow across the domain. This decrease in surface wind speed could decrease the advection of warm, moist air in the lower levels. This increase in surface friction may affect

MCS maintenance on a finer scale not resolved at 50km. The 850 mb wind speeds from the Gulf of Mexico decreased in each case in the 1850-vegetation experiments by up to 5 m/s. This decrease in 850 mb wind speed is likely a consequence of the decreased pressure gradients (height gradients).

In all of the cases, the area average CAPE increased in the 1850-vegetation experiments between 5-15%. The area average CIN decreased in all of our cases between 5-10%. Although the area average CAPE increased and CIN decreased, the maximum CAPE and CIN in our cases did not appear to follow a pattern as clearly as the area average CAPE and CIN. Even in the morning May 6, 2008 case (not shown in figures), the maximum CAPE was lower in the 1850-vegetation experiment and maximum CIN was lower. One might think that this decrease would help to decrease precipitation in the 1850-vegetation experiment, however the May 6 case showed increased overall precipitation across Iowa in the 1850-vegetation experiment. There is no clear connection between the maximum CAPE and CIN in these events, however previous work suggests higher values of CIN may be important for heavier precipitation, as the buoyant air can be concentrated in a smaller area similar to backbuilding of quasi-stationary convection (Junker and Schneider, 1997).

One other case (June 8, 1993) was an example of failure to trigger convection, where the reflectivity (diagnosed from WRF output using the WRF-python package) from the model depicted convection over Wisconsin, but the model output precipitation data did not show that rainfall had occurred.

In summary, the case studies revealed a consistent decrease in precipitation in response to the 1850-vegetation experiments, often associated with a decrease in the area of heavy precipitation. Only the May 6, 2008 case showed an increase in precipitation with 1850

vegetation, but this was a morning MCS where CAPE was not very high in either experiment and surface heating had not yet occurred. The rainfall rates in the case studies were also consistent with the probability density plots showing an increase in moderate to heavy precipitation in the present-day vegetation experiment.

2.4 Conclusions

In summary, we simulated the effects of historical land use and land cover changes on rainfall associated with two major flood events in the US Midwest. The prescribed 1850-vegetation experiment increased vegetation cover (ELAI) in the United States, especially in the central, southern, and eastern portions of the US. Forests and woody savannas replaced croplands and less vegetated areas in the 1850-vegetation experiment compared to the present-day vegetation experiment. This lowered the albedo due to forests and woody savannas being darker in color, and it also changed (tending to reduce) the surface latent heat fluxes.

The results from the 1993 and 2008 simulations generally agreed. The changes in albedo and transpiration appeared to dominate the precipitation response to 1850 vegetation. The change in transpiration had effects on a local scale, and the changes in albedo ultimately contributed to changes in the large scale circulations. Precipitable water and latent heat flux change have similar changes which led to a decrease in precipitation. The simulations capture the observed timing and location of precipitation during the flood events because of the lateral boundary conditions feeding the synoptic troughs into the model.

The 1850 vegetation led to decreased precipitation in our simulations. Large changes in albedo across the eastern United States likely led to changes in latent and sensible heat flux, where the latent heat flux decreased and the sensible heat flux increased. This increased the

surface temperature and decreased evaporation at the surface. The changes led to large-scale changes in the pressure gradients, weakening the LLJ. A weakened LLJ will transport less moisture and warm air needed for MCS. The weakened LLJ combined with the decreased latent heat flux gave us a subtle yet noticeable difference in precipitable water with 1850 vegetation. The decrease in pressure over the central and eastern US could have larger effects on the large-scale circulations suggesting that the change in land-use could be affecting the climate on a larger scale. The decrease in latent heat flux in 1850 may have contributed to the decrease in 2-meter water vapor mixing ratio and precipitable water.

Previous studies focused on mean precipitation changes from land use land cover change in the United States (Chen and Dirmeyer 2017; Devanand et al., 2020). Our study investigates the land use land cover change effects on specific MCS events and their extreme precipitation amounts. We found that the precipitation in our model generally remained in the same location from the present-day vegetation to the 1850 vegetation experiment. The decrease in average precipitation across the domain is much more apparent than the change in individual events. The land use change often caused a decrease in precipitation overall, but the precipitation maxima remained or weakened very slightly. This decrease in precipitation during these events could lead to less severe flooding across the Midwest. However, previous work found that coarser model resolution led to weaker rainfall while finer resolution increases precipitation amounts (Devanand et al., 2020). Future work can investigate these flood events with a finer resolution to see if this also occurs in the 1993 and 2008 flood cases and whether or not we still see a decrease in precipitation in the 1850 vegetation experiments.

The surface fluxes significantly changed in response to the 1850 vegetation experiment. Sensible heat fluxes increased by 70-130 W/m² and latent heat flux decreased by up to 60-90

W/m² in some areas, especially in western Illinois. The responses in the 2008 event were similar but slightly less in magnitude than the 1993 event. This is likely because the 2008 event began earlier in the spring than the 1993 event when crops had not reached peak leaf area. Crop evapotranspiration was typically higher than that of forests, which led to the muted spring response to the 1850 LULC change.

Although there was a decrease in precipitation with the 1850-vegetation, the accumulated precipitation minus evaporation (P-E) remained relatively unchanged by the end of the event periods. The lower evaporation rates with 1850-vegetation meant that less water at the surface was evaporating, which offset the decrease in precipitation in this experiment. Therefore, the decrease in precipitation with forests in the 1850 vegetation scenario does not necessarily imply less flooding since the supply of water at the surface (P-E) is relatively unchanged by the vegetation.

The model struggled to accurately predict the rainfall magnitudes. The present-day vegetation simulations underpredicted the total rainfall by 64% in the 1993 simulations and 74% in the 2008 simulations. The cumulative area-averaged precipitation in the 1850-vegetation simulations decreased by 16% in the 1993 experiments from 153 mm to 129 mm and 13% in the 2008 simulations from 153 mm to 130 mm. The Betts-Miller-Janjic (BMJ) convective scheme is triggered by CAPE. The convective scheme is likely not the only factor that led to decreased precipitation totals in the present-day vegetation simulations. Lower resolution can be a problem for precipitation totals because the model cannot explicitly resolve convection. Oftentimes, the model failed to trigger convection at all. Higher values of reflectivity are not resolved at 50 km resolution because these usually do not exist on a scale of 50 km. In our model, this sensitivity to CAPE in the convective scheme could be a reason for the low bias in precipitation. Precipitation

errors could also be due to the microphysics scheme. The 50 km simulations also struggled to capture smaller scale processes that affect MCSs. Running these simulations at a finer resolution would tell us much more about the effects of land-use land cover change on MCS structure.

There was a decrease in CAPE in the 1850-vegetation simulations, which is physically consistent with the decrease in precipitation in our results. By analyzing the 24 hours leading up to the event, it was found that the area averaged CAPE was 5-15% larger on average in the 1850-vegetation simulations than the present-day vegetation simulations. The CIN was also 5-15% higher in the 1850-vegetation simulations. This change in average CAPE and CIN will certainly affect a convective scheme sensitive to CAPE changes.

With the 1850 vegetation we found there was a decrease in precipitable water. This decrease helps explain the decrease in precipitation in the 1850 vegetation experiment, but by itself is not enough to explain the full magnitude of changes in precipitation. These results suggest that dynamical and CAPE changes are also playing a role in the precipitation decrease in the 1850-vegetation experiments. The surface fluxes are not only changing the local environment, but large scale circulations. Weaker 850 mb winds in the 1850-vegetation experiment could cause weakened convergence and upward motion at the front, weakening the system relative to the present-day vegetation experiment.

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CHAPTER 3. GENERAL CONCLUSIONS

Over the 1993 and 2008 study periods, the prescribed 1850 LULC increased vegetation amount over much of the US Midwest and Southeast. Using an effective leaf area index (ELAI) to quantify the combined effects of the fractional cover of each vegetation type (FVEG) and its leaf area index, the ELAI was greater in 1850 primarily due to forests existing where there is crop cover in the present-day. This led to an increase in sensible heat flux and a decrease in latent heat flux. The latent heat flux decreased because the crops transpire more than the forests. The sensible heat flux most likely increased because of a decrease in albedo in the 1850-vegetation simulations.

The changes in the surface heat fluxes in the central US changed the local environment and the large-scale circulations in our model domain. The increased sensible heat flux in the 1850-vegetation experiment increased the surface air temperature in the central US. The latent heat flux decreased, leading to less moisture at the surface since savannas and forests transpire less than the crops in the present-day experiment. These thermodynamic changes led to decreased pressure level heights in the 1850-vegetation experiment across the central US. This decrease in pressure level heights in our simulations led to a decrease in pressure gradients across the domain. The decreased pressure gradient led to weaker low-level winds (e.g. 850 mb). These decreases in moisture and changes in large scale circulations decreased the precipitation in the 1850-vegetation experiments during the 1993 and 2008 flood events.

Precipitation changes were apparent in both the 1993 and 2008 simulations. The 1850-simulations had decreased overall precipitation in the Midwest. We found that with each event, the average precipitation across the domain was reduced slightly in the 1850-vegetation simulations, but the precipitation maxima often remained similar in magnitude. The probability

density plots (Figure 2.14, 2.23) revealed that the present-day vegetation experiment produced more precipitation in the moderate rainfall rates (2-12 mm/hr) in both experiments. There was not as clear of a change at extreme rainfall rates over 15 mm/hr in the 1993 event. For the 2008 event, the present-day vegetation experiment had greater frequency of extreme precipitation values of 15 mm/hr compared to the 1850-vegetation experiment. In summary, our findings show that the precipitation remained in the same general locations in both the present-day and 1850-vegetation experiments, but the 1850-vegetation resulted in lower precipitation totals. One interesting finding was that the reduction in precipitation came about more from a reduction in the area of moderate to heavy rain rather than a reduction in the highest rain rates.

Although there was a decrease in precipitation with the 1850-vegetation experiments, evaporation also decreased so that the change in precipitation minus evaporation (P-E) was not very significant. The flooding with 1850-vegetation would likely not be reduced despite the decrease in precipitation because the surface evaporation was lower.

The next steps for this project are to run higher resolution convective-allowing simulations. Low resolution simulations such as climate models and the 50-km resolution used in this study require a convective scheme. Various convective schemes could be tested with these cases to see if results align with those in this study. This study could also be repeated with microphysics schemes that are more efficient at producing rainfall to better match the observed rainfall rates. A more efficient microphysics scheme may give more insight into the changes in extreme precipitation maxima. Future work could also investigate other extremes such as droughts. It would be interesting to see if LULC change reduces the extremes in both cases, and why that might be. It may be worthwhile to study the effects of LULC change on a drought year, a typical year, or over a longer period of time. Running the model in drier years could give

insight on whether or not the same results occur from the land use land cover change with less moisture present, and running the model for a longer time period may reveal worsening trends over time.